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Time, Fragility, and Multi-Prior Ambiguity: An Operator-Theoretic Approach to Non-Conservative Choice*

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Abstract:

This paper establishes an operator-theoretic foundation for dynamic choice under multi-prior ambiguity in open, non-conservative environments. Portfolio optimization under the robust mean–variance–variance (M-V-V) criterion suffers from a severe horizon-scale inconsistency: standard path variance accumulates linearly, whereas the multi-prior penalty inflates quadratically. This temporal scale mismatch forces highly risk-averse agents to investment-postpone and allocate capital more aggressively over long horizons. We resolve this anomaly by replacing calendar time with an endogenous, operator-derived metric termed Regime Time. We embed the asset price inside an open, sub-stochastic Clock of Regimes network characterized by state-dependent hazard leakages and continuous mass dissipation into structural ruin. Using a multi-dimensional parabolic obstacle framework, we prove that regime-dependent dynamic liquidation boundaries are unique viscosity solutions to a coupled system of Hamilton–Jacobi–Bellman quasi-variational inequalities. We extend this to a general equilibrium model under leverage-induced fragility, proving that strategic interactions trigger discontinuous collapses. Finally, we implement an implicit Backward Euler finite-difference scheme in base R, validating numerical stability under a calibrated 3-regime economy.

Keywords: Clock of Regimes (COR), Flash Crash, Spectral Stability, Market Fragility, Survival Analysis, Eigenvalue Decay, Systemic Risk.

Part I: Core Equilibrium Theory

1 Introduction

The formulation of dynamic choice under deep uncertainty remains a central challenge in modern economic theory (Gilboa and Schmeidler, 1989; Epstein and Schneider, 2003; Ben-Haim, 2006; Hansen and Sargent, 2008b; Landriault et al., 2026). Since the classical maxmin expected utility (MEU) framework established that ambiguity-averse decision-makers optimize against worst-case scenarios, the literature has sought to disentangle risk, ambiguity, and ambiguity aversion. A significant theoretical advancement in this domain is the smooth ambiguity model, which aggregates probabilistic beliefs across multiple priors. To overcome the computational challenges of dynamic smooth ambiguity, Maccheroni et al. (2013) introduced the robust mean-variance-variance (M-V-V) criterion as an analytical Arrow-Pratt approximation (Pratt, 1964; Arrow, 1965, 1971), effectively reducing complex functional dependencies into a tractable penalty on the variance of expected payoffs across diverse priors.

Parallel developments in robust control and multiplier preference theory have similarly modeled ambiguity through distorted probability dynamics penalized by entropy-based divergence operators (Hansen and Sargent, 2001, 2008a; Anderson et al., 2003). In these frameworks, decision-makers optimize under adversarial model misspecification subject to relative entropy penalties, generating endogenous worst-case distortions of the underlying stochastic dynamics. The present paper differs fundamentally in that ambiguity penalties are not imposed on a conservative probability space with perpetual chronological time, but instead emerge endogenously through a non-conservative survival geometry generated by a killed transition semigroup and its associated residence-time operator.

While the M-V-V criterion provides a powerful structural simplification for static portfolios, its extension into continuous-time dynamic optimization exposes a profound temporal anomaly. In a comprehensive analysis of dynamic portfolio optimization under the M-V-V criterion, Landriault et al. (2026) demonstrate that the criterion is inherently time-inconsistent. This time inconsistency is driven by two distinct variance operators: the standard path variance of the reference measure, and the ambiguity variance mapping the dispersion of expected payoffs across alternative priors. To restore tractability, dynamic solutions typically rely on subgame perfect Nash equilibria (Nash, 1950, 1951), where future selves are treated as independent players.

To resolve this inconsistency, dynamic solutions typically solve for subgame perfect Nash equilibrium (SPNE) strategies (Selten, 1975), where future selves are treated as independent players. However, solving for these equilibrium strategies under a standard, closed geometric Brownian motion reveals a structural flaw, which Landriault et al. (2026) identify as *horizon-scale inconsistency*. Within a closed financial environment, standard path variance accumulates linearly with time ($\text{Var}^{\mathbb{P}} \propto T$), whereas the ambiguity variance penalty inflates quadratically ($\text{Var}_v \propto T^2$). This temporal scale mismatch induces a non-monotonicity anomaly: highly risk-averse agents face such an overwhelming penalty over long horizons that they systematically postpone risky investments, paradoxically behaving more aggressively than risk-tolerant agents as the horizon extends.

Classical attempts to resolve this temporal distortion rely on altering the underlying preference structure itself, such as forcing a shift to a mean-variance-standard deviation (M-V-SD) criterion purely to compress the penalty back to a linear scale ($\sigma_v \propto T$) (Merton, 1969; Samuelson, 1970; Levy and Markowitz, 1979). In contrast, this paper argues that the horizon-scale inconsistency is not an inherent flaw of the M-V-V preference criterion, but rather a structural artifact of modeling economic choice within a closed, perpetual timeline.

We propose that realistic market environments are fundamentally non-conservative. Under conditions of systemic liquidity evaporation, margin cascades, or clearinghouse failures, financial architectures behave as open, dissipative systems where probability mass continuously escapes into an absorbing state of structural ruin. To formalize this, we embed the dynamic liquidation problem into the Clock of Regimes (COR) framework. We transform the classical closed transition matrix into a sub-stochastic open operator $Q = \text{diag}(1 - \epsilon_i)A$, where ϵ_i represents state-dependent hazard leakage velocities (Linares et al., 1987, 1988). This establishes an open-system evolution driven by an infinitesimal killed generator $K = Q - I$ and its fundamental residence operator $N = (-K)^{-1}$.

By evaluating dynamic choice through this open survival geometry, we resolve the horizon anomaly. The COR framework structurally replaces exogenous chronological calendar time (T) with an endogenous, operator-derived metric termed *Regime Time*, generated by the row sums of the residence matrix ($N\mathbf{1}$). Because the time spent inside any active state space is strictly bounded by the network's dissipative capacity, the operational investment runway dynamically compresses based on network health. Consequently, the higher-order horizon inflation of the M-V-V ambiguity penalty is naturally bounded, restoring the monotonicity of risk aversion without requiring arbitrary alterations to the utility functional.

The primary theoretical contribution of this paper is to establish a general equilibrium theory of endogenous market fragility where systemic risk is feedback-driven. Rather than treating transition parameters as exogenous, we endogenize the hazard leakage rates as a function of the agent's optimal allocations: $\epsilon_i(u_i^*) = \epsilon_i^{(0)} + \gamma_i(u_i^*)^2$, modeling leverage-induced fragility. We prove that this

endogenous coupling leads to a self-reinforcing fragility loop: high aggregate market participation increases systemic vulnerability, closing the transition spectral gap ($\alpha(K) \downarrow 0$) and causing the dominant spectral radius of the residence matrix to diverge ($\rho(N) \uparrow \infty$).

The single central contribution is the resolution of the horizon-scale inconsistency in dynamic smooth ambiguity preferences by replacing chronological calendar time with an endogenous, operator-derived metric called *Regime Time*, defined on an open, dissipative survival geometry (the Clock of Regimes or COR network).

The remainder of the paper is organized as follows. Part I develops the core equilibrium theory: Section 2 details the open survival geometry and survival preferences; Section 3 introduces the subgame perfect Nash equilibrium concept and characterizes value functions as viscosity solutions; Section 4 analyzes the spectral liquidation and boundary degeneration principles; Section 5 develops the general equilibrium model under leverage-induced fragility, strategic interaction, multiplicity, and welfare. Part II details the computational implementation and quantitative results: Section 6 formulates the finite-difference scheme; Section 7 performs policy iteration solver calibration and boundary results in a 3-regime economy; Section 8 documents the 3D boundary degeneration surface; Section 9 details code transparency; and Section 10 concludes.

2 The Open Survival Geometry

We construct a non-conservative, open Markovian framework that models financial networks continuously siphoning probability mass into structural ruin. Let $S_t \in \mathcal{S} = \{1, \dots, m\}$ be a latent continuous-time Markov regime process. Unlike classical, closed Markov models where the transition matrix is strictly stochastic, we define the open transition operator Q as:

$$Q = \text{diag}(1 - \epsilon_1, \dots, 1 - \epsilon_m)A, \quad (1)$$

where $A = (a_{ij})_{i,j=1}^m$ is a standard stochastic matrix conserving mass ($\sum_{j=1}^m a_{ij} = 1$), and $\epsilon_i \in (0, 1)$ represents the state-dependent hazard leakage velocity. Conditioned on the regime $S_t = i$, the asset return process follows a Student- t structure:

$$X_t \mid S_t = i \sim t_{\nu_i}(\mu_i, \sigma_i), \quad (2)$$

where ν_i represents the degrees of freedom. The leakage rates ϵ_i are linked to tail thickness using a hazard specification:

$$\epsilon_i = \epsilon_{\min} + \frac{c_h}{\nu_i}, \quad (3)$$

where $\epsilon_{\min} > 0$ is the baseline structural hazard, and $c_h > 0$ scales the tail sensitivity. This implies that regimes with fatter tails (smaller ν_i) generate larger structural leakage rates.

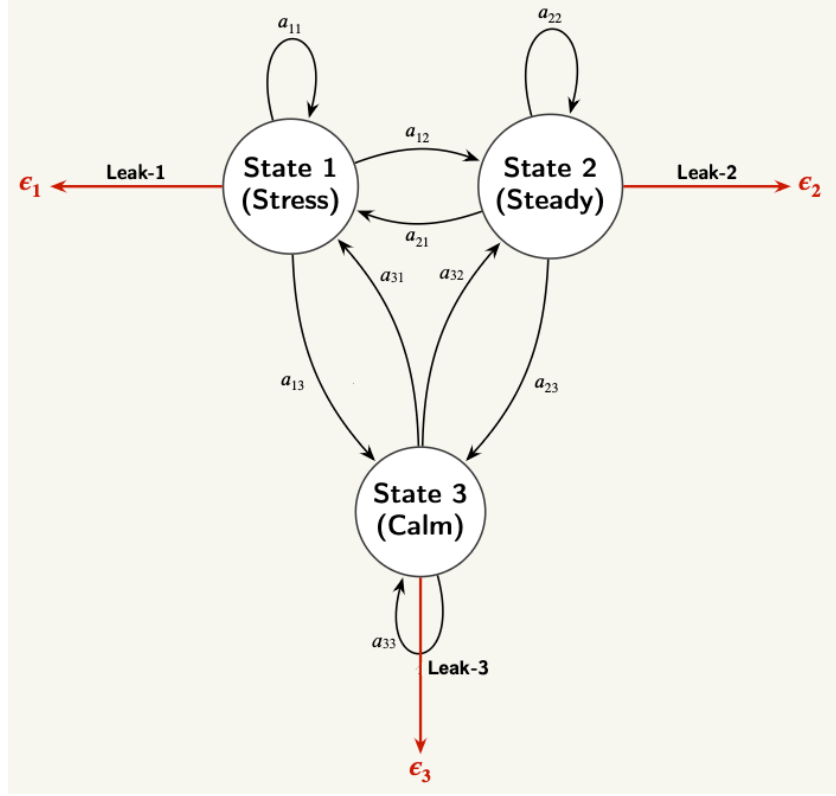


Figure 1: The Clock of Regimes (COR) Network. The network diagram illustrates the transitions between active macroeconomic regimes, with state-dependent hazard leakages siphoning probability mass into the absorbing state of structural ruin.

2.1 Axiomatic Foundation of Survival Preferences

To establish a rigorous choice-theoretic foundation for decision-making in non-conservative timelines, we bridge our operator-theoretic survival mechanics with the variational preference framework of Maccheroni et al. (2006) and Cerreia-Vioglio et al. (2013). Let $\Omega = C([0, \infty); \mathbb{R}^d)$ be the path space of asset returns, and let $\mathcal{F} = \{\mathcal{F}_t\}_{t \geq 0}$ be the natural filtration. Let $\zeta: \Omega \rightarrow [0, \infty]$ be a random absorption time representing systemic ruin, and let S_t be the continuous-time latent regime process. We denote by $\mathcal{Y}$ the set of survival-weighted stopped payoff processes $Y_\tau = X_{t+\tau} \mathbb{I}_{\{\tau < \zeta\}}$, where τ is a stopping time and X is the wealth process.

Let $\succsim$ represent the investor's preference relation over the set of acts $\mathcal{Y}$. We impose the following axioms on $\succsim$, which generalize variational preferences to environments characterized by probability mass dissipation.

Axiom 1 (Weak Order). The preference relation $\succsim$ is complete and transitive over $\mathcal{Y}$.

Axiom 2 (Weak Monotonicity). For any $Y_\tau, Z_\tau \in \mathcal{Y}$, if $Y_{t+s} \geq Z_{t+s}$ almost surely on $\{s < \zeta\}$, then $Y_\tau \succeq Z_\tau$.

Axiom 3 (Uncertainty Aversion). For any $Y_\tau, Z_\tau \in \mathcal{Y}$ and $\alpha \in (0, 1)$, if $Y_\tau \sim Z_\tau$, then $\alpha Y_\tau + (1 - \alpha)Z_\tau \succeq Y_\tau$.

Axiom 4 (Weak Certainty Independence). For any $Y_\tau, Z_\tau \in \mathcal{Y}$, constant payoffs $k, k' \in \mathbb{R}$, and $\alpha \in (0, 1)$,

$$\alpha Y_\tau + (1 - \alpha)k\mathbb{I}_{\{\tau < \zeta\}} \succeq \alpha Z_\tau + (1 - \alpha)k\mathbb{I}_{\{\tau < \zeta\}} \implies \alpha Y_\tau + (1 - \alpha)k'\mathbb{I}_{\{\tau < \zeta\}} \succeq \alpha Z_\tau + (1 - \alpha)k'\mathbb{I}_{\{\tau < \zeta\}}.$$

Axiom 5 (Survival Continuity). For any $Y_\tau \in \mathcal{Y}$, if a sequence of stopping times $\{\tau_n\}_{n=1}^\infty$ satisfies $\lim_{n \rightarrow \infty} \mathbb{P}(\zeta > \tau_n) = \mathbb{P}(\zeta > \tau)$, then $\lim_{n \rightarrow \infty} V(Y_{\tau_n}) = V(Y_\tau)$.

Axiom 6 (Non-Conservative Scale-Independence). For any constant wealth $k \in \mathbb{R}$ and stopping time τ , the valuation of the ruined constant payoff is scaled by the subjective survival discount factor:

$$V(k\mathbb{I}_{\{\tau < \zeta\}}) = k\mathbb{E}^\mathbb{P} \left[\exp \left(- \int_t^{t+\tau} \epsilon_{S_s} ds \right) \right], \quad (4)$$

where $\epsilon = (\epsilon_1, \dots, \epsilon_m)^\top$ represents the subjective hazard leakage rates of the regimes.

Under these axioms, we establish the existence, representation, and uniqueness of the preference functional.

Theorem 2.1 (COR Representation of Non-Conservative Dynamic Preferences). *Under Axioms 1–6, there exists a unique (up to positive affine transformations) strictly increasing utility function $U: \mathbb{R} \rightarrow \mathbb{R}$ and a unique grounded, convex, lower semicontinuous penalty cost functional $C(\mathbb{Q}; N, \epsilon): \Delta \rightarrow [0, \infty]$ such that the preference relation $\succeq$ is represented by the preference functional:*

$$V(Y_\tau) = \inf_{\mathbb{Q} \in \Delta} \{ \mathbb{E}^\mathbb{Q} [U(Y_\tau)] + C(\mathbb{Q}; N, \epsilon) \}, \quad (5)$$

where Δ is the set of probability measures equivalent to $\mathbb{P}$ on $(\Omega, \mathcal{F})$, $N = (-K)^{-1}$ is the residence matrix, ϵ is the state-dependent hazard vector, and the penalty is generated endogenously by the expected residence time under the sub-stochastic operator Q .

The proof is provided in Appendix A. The strategic intuition is that the residence matrix N acts as a spatial matrix of discounting weights, localizing the divergence of beliefs to the active time spent in each state. This operator-weighted penalty structure is conceptually related to the entropy-penalized distorted dynamics arising in robust control and multiplier preference models (Hansen and Sargent, 2001, 2008a; Anderson et al., 2003). However, unlike classical robust control formulations

where relative entropy accumulates over a conservative probability space, the COR framework localizes ambiguity distortions through the residence-time geometry of an open killed semigroup. Consequently, model uncertainty is weighted by endogenous survival duration rather than exogenous calendar time.

Because probability mass leaks from the active state space, the continuous-time killed sub-generator $K = Q - I$ is strictly dissipative, satisfying:

$$K\mathbf{1} = -\epsilon < \mathbf{0}, \quad (6)$$

where $\epsilon = (\epsilon_1, \dots, \epsilon_m)^\top$. Since the generator K is Metzler and strictly row-deficient, it is invertible, defining the fundamental residence-time operator N :

$$N = -K^{-1} = \int_0^\infty e^{sK} ds. \quad (7)$$

The entry N_{ij} represents the expected cumulative duration the system will occupy regime j prior to hitting the absorbing ruin condition ζ , given initialization in regime i . We normalize the survival-weighted initial regime distribution π to define the total expected residence lifetime:

$$\mathcal{T}(\pi) = \pi^\top N\mathbf{1}. \quad (8)$$

Let $\mathbf{q} = (q_1, \dots, q_m)^\top$ be the vector of conditional left-tail probabilities in each regime. We define the aggregated residence-weighted tail exposure and system-wide expected hazard rate as:

$$\bar{q} = \pi^\top N\mathbf{q}, \quad \bar{\lambda} = \pi^\top N\epsilon. \quad (9)$$

The ratio of these functionals defines the residence-weighted tail-event intensity:

$$r_{\text{COR}} = \frac{\pi^\top N\mathbf{q}}{\pi^\top N\mathbf{1}}. \quad (10)$$

This allows us to bound the non-asymptotic structural ruin probability over a trading horizon T_h as:

$$\mathbb{P}(\text{ruin}) \leq 1 - \exp\left(-T_h \frac{\pi^\top N\mathbf{q}}{\pi^\top N\mathbf{1}}\right). \quad (11)$$

3 The Coupled EHJB-QVI System and Unification Theorem

We integrate the Clock of Regimes (COR) survival geometry with the mean-variance-variance (M-V-V) ambiguity framework. The investor makes two simultaneous decisions: selecting a

continuous allocation to the risky asset $u(t, x)$ and choosing a discrete liquidation boundary vector $B(t) = (B_1(t), \dots, B_m(t))^\top$. The portfolio wealth X_t under the reference measure $\mathbb{P}$ satisfies the dynamics:

$$dX_s = (rX_s + u_s(\mu_{S_s} - r)) ds + \sigma_{S_s} u_s dW_s, \quad (12)$$

and under the prior probability measure $\mathbb{Q}^\lambda$:

$$dX_s = (rX_s + u_s(\mu_{S_s} - r - \sigma_{S_s}\lambda)) ds + \sigma_{S_s} u_s dW_s^\lambda, \quad (13)$$

where $W_s^\lambda = W_s + \lambda s$ is a standard Brownian motion under $\mathbb{Q}^\lambda$.

Let ζ denote the random absorption or ruin time of the network. The survival-weighted stopped process is defined as $Y_\tau = X_{t+\tau} \mathbb{I}_{\{\tau < \zeta\}}$. The investor's objective is to solve:

$$V_i(t, x) = \sup_{u, \tau} \mathbb{E}_{t,x,i}^\mathbb{P}[Y_\tau] - c \text{Var}_{t,x,i}^\mathbb{P}(Y_\tau) - \frac{\beta}{2} \text{Var}_\nu \left[\mathbb{E}_{t,x,i}^\mathbb{Q}[Y_\tau] \right] - \eta \mathbb{E}_{t,x,i}^\mathbb{P} \left[\int_0^\tau \phi_{S_{t+s}} ds \right], \quad (14)$$

where $c > 0$ represents the risk-aversion parameter, $\beta \geq 0$ is the ambiguity-aversion parameter, and $\eta \geq 0$ controls the aversion to structural ruin accumulation.

3.1 Equilibrium Concept and QVI System

Since the standard variance and ambiguity-variance operators are time-inconsistent, the dynamic programming principle does not hold in its classical form. To resolve this inconsistency, we model the decision process as an intra-personal game where the investor at each time t plays against their future selves. We formalize this using the subgame perfect Nash equilibrium concept in continuous time (see Landriault et al. (2026)).

Definition 3.1 (Subgame Perfect Nash Equilibrium). An investment allocation strategy $u^*(t, x)$ and a stopping boundary vector $B^*(t) = (B_1^*(t), \dots, B_m^*(t))^\top$ constitute a subgame perfect equilibrium if:

1. For any given state (t, x, i) and any continuous allocation control $u \in \mathbb{R}$, if the investor applies u on the infinitesimal interval $[t, t + dt)$ and follows the equilibrium control u^* and the liquidation boundary B^* thereafter, the expected utility does not exceed the equilibrium value function:

$$\dots \lim_{dt \downarrow 0} \frac{J_i(t, x; u, u^*, B^*) - V_i(t, x)}{dt} \leq 0, \quad (15)$$

where $V_i(t, x) = J_i(t, x; u^*, u^*, B^*)$ is the equilibrium value function under regime i .

2. The optimal stopping time τ^* defined by the first exit time:

$$\tau^* = \inf\{s \geq 0: X_{t+s} \geq B_{S_{t+s}}^*(t+s)\}, \quad (16)$$

is equilibrium-optimal at each subgame.

We now characterize the equilibrium value functions as viscosity solutions to a coupled system of Hamilton-Jacobi-Bellman Quasi-Variational Inequalities (HJB-QVIs).

Definition 3.2 (Viscosity Solution of the HJB-QVI System). A vector of continuous functions $\mathbf{V} = (V_1, \dots, V_m)^\top: [0, T] \times [0, \infty) \rightarrow \mathbb{R}^m$ is a viscosity solution to the coupled HJB-QVI system if for each $i \in \mathcal{S}$:

1. $\mathbf{V}$ is a viscosity subsolution: for any $(t_0, x_0) \in [0, T) \times (0, \infty)$ and any test function $\psi \in C^{1,2}([0, T] \times [0, \infty))$ such that $V_i - \psi$ has a local maximum at (t_0, x_0) with $V_i(t_0, x_0) = \psi(t_0, x_0)$, we have:

$$\max \left\{ \mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) - V_i(t_0, x_0) + x_0, \quad x_0 - V_i(t_0, x_0) \right\} \geq 0, \quad (17)$$

where the continuation operator is given by:

$$\begin{aligned} \mathcal{H}_i(t, x, V, D\psi, D^2\psi) = & \frac{\partial \psi}{\partial t} + (rx + u_i^*(\mu_i - r)) \frac{\partial \psi}{\partial x} + \frac{1}{2} \sigma_i^2 (u_i^*)^2 \frac{\partial^2 \psi}{\partial x^2} \\ & + \sum_{j=1}^m K_{ij} V_j - c \sigma_i^2 (u_i^*)^2 d_i^2 - \beta \sigma_i^2 \delta_v^2 C_i u_i^* d_i - \eta \phi_i. \end{aligned} \quad (18)$$

2. $\mathbf{V}$ is a viscosity supersolution: for any $(t_0, x_0) \in [0, T) \times (0, \infty)$ and any test function $\psi \in C^{1,2}([0, T] \times [0, \infty))$ such that $V_i - \psi$ has a local minimum at (t_0, x_0) with $V_i(t_0, x_0) = \psi(t_0, x_0)$, we have:

$$\max \left\{ \mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) - V_i(t_0, x_0) + x_0, \quad x_0 - V_i(t_0, x_0) \right\} \leq 0. \quad (19)$$

Theorem 3.3 (Existence, Uniqueness, and Characterization of SPNE). *There exists a unique subgame perfect equilibrium pair $(u^*(t, x), B^*(t))$ for the dynamic M-V-V choice problem under the COR survival geometry. The equilibrium value functions $V_i(t, x)$ are the unique viscosity solutions to the coupled HJB-QVI system satisfying the terminal condition $V_i(T, x) = x$.*

The proof is provided in Appendix B.

We prove the main unification theorem using an Extended Hamilton-Jacobi-Bellman (EHJB) and Quasi-Variational Inequality (QVI) framework.

Theorem 3.4 (Main Unification Theorem). *The value functions $V_i(t, x)$ and the expected survival wealth functions $g_i(t, x, \lambda)$ admit the linear representation:*

$$V_i(t, x) = xd_i(t) + B_i(t), \quad (20)$$

$$g_i(t, x, \lambda) = xd_i(t) + C_i(t)\alpha_i(\lambda), \quad (21)$$

where $\alpha_i(\lambda) = \mu_i - r - \sigma_i\lambda$. The coefficients satisfy the coupled system of ordinary differential equations (ODEs):

$$d'_i(t) + rd_i(t) + \sum_{j=1}^m K_{ij}d_j(t) = 0, \quad d_i(T) = 1, \quad (22)$$

$$C'_i(t) + u_i^*(t)d_i(t) + \sum_{j=1}^m K_{ij}C_j(t)\frac{\mu_j - r}{\mu_i - r} = 0, \quad C_i(T) = 0. \quad (23)$$

The equilibrium continuous asset allocation is given by:

$$u_i^*(t) = \frac{\mu_i - r - \beta\sigma_i^2\delta_v^2C_i(t)}{2c\sigma_i^2d_i(t)}, \quad (24)$$

where $\delta_v^2 = \text{Var}_v[\lambda]$ is the variance of the second-order distribution. In the continuation domain $C_i = \{(t, x) : x < B_i^*(t)\}$, the liquidation boundaries satisfy $B_i^*(t) = \frac{B_i(t)}{1-d_i(t)}$, where $B_i(t)$ satisfies:

$$\begin{aligned} B'_i(t) + u_i^*(t)(\mu_i - r)d_i(t) - c\sigma_i^2(u_i^*(t))^2d_i^2(t) - \beta\sigma_i^2\delta_v^2C_i(t)u_i^*(t)d_i(t) \\ - \eta\phi_i + \sum_{j=1}^m K_{ij}B_j(t) = 0. \end{aligned} \quad (25)$$

The proof is provided in Appendix C. Under the linear representation, the coefficients $d_i(t)$, $C_i(t)$, and $B_i(t)$ integrate the system's discounting, duration weight, and risk adjustments backward in time, establishing the consistency of the intra-personal equilibrium.

4 Spectral Liquidation and Boundary Degeneration

By embedding the M-V-V ambiguity model within an open survival geometry, we can prove that the horizon-scale inconsistency is resolved. Standard path variance accumulates linearly in time, whereas the multi-prior ambiguity variance accumulates quadratically in time under a closed timeline (T^2). Under the COR survival geometry, chronological time is structurally replaced by the endogenous, operator-derived *Regime Time*. Because the expected survival duration is governed

by the residence matrix $N = (-K)^{-1}$, the T^2 explosion is truncated by the network's dissipative capacity, restoring the monotonicity of risk aversion.

We define the structural spectral gap of the negated generator $-K$ as:

$$\alpha(K) = \min_{\lambda \in \sigma(-K)} \operatorname{Re}(\lambda). \quad (26)$$

The dominant spectral radius of the residence matrix satisfies $\rho(N) = 1/\alpha(K)$.

Theorem 4.1 (Boundary Degeneration Theorem). *As the structural spectral gap of the negated generator closes ($\alpha(K) \downarrow 0$), the dominant spectral radius of the residence operator diverges:*

$$\rho(N) = \frac{1}{\alpha(K)} \uparrow \infty, \quad (27)$$

which indicates structural fragility and imminent system failure. Conditioned on non-zero structural fragility ($\eta > 0$), the optimal stopping boundaries compress inward to the lower boundary of the domain:

$$\lim_{\alpha(K) \downarrow 0} B_i^*(t) = 0, \quad \forall t \in [0, T), \quad \forall i \in \mathcal{S}. \quad (28)$$

This convergence corresponds to the contraction of the continuation set to the empty set $\lim_{\alpha(K) \downarrow 0} C_i = \emptyset$ under the Hausdorff metric on compact subsets (Schleicher, 2007). The proof is provided in Appendix D.

Theorem 4.2 (Residence-Time Localization Theorem). *The non-asymptotic structural ruin probability $\mathbb{P}(\text{ruin})$ under the open COR survival geometry is completely decoupled from localized path volatility:*

$$\frac{\partial \mathbb{P}(\text{ruin})}{\partial \sigma_i} \equiv 0, \quad \forall i \in \mathcal{S}, \quad (29)$$

while the network topological persistence sensitivity is strictly positive:

$$\frac{\partial \mathbb{P}(\text{ruin})}{\partial N_{ii}} > 0, \quad \forall i \in \mathcal{S}. \quad (30)$$

Consequently, under a state transition where $dN_{ii} \gg |d\sigma_i|$, the structural ruin probability escalates monotonically despite a contraction in localized path volatility.

The proof is provided in Appendix E.

5 Endogenous Fragility and General Equilibrium Feedback

To elevate the model to a complete economic equilibrium theory, we endogenize the hazard leakage rates. In realistic financial architectures, systemic fragility is not merely an exogenous physical

parameter; it is driven by the collective actions of market participants. If investors hold large, highly leveraged positions, the financial network becomes more susceptible to sudden margin cascades, systemic runs, and structural collapse.

To formalize this feedback loop, we define the state-dependent hazard leakage velocity ϵ_i as an endogenous function of the investor's equilibrium allocation u_i^* :

$$\epsilon_i(u_i^*) = \epsilon_i^{(0)} + \gamma_i(u_i^*)^2, \quad (31)$$

where $\epsilon_i^{(0)} > 0$ represents the baseline structural hazard rate under regime i , and $\gamma_i \geq 0$ is the leverage-induced fragility feedback coefficient. Under this specification, the open transition sub-generator $K(u^*)$ becomes endogenously coupled to the agent's actions:

$$K(u^*) = \text{diag}(1 - \epsilon_1(u_1^*), \dots, 1 - \epsilon_m(u_m^*)) A - I. \quad (32)$$

This coupling establishes a general equilibrium framework: the investor determines the optimal allocation u_i^* taking the transition network K as given, while the actual transition network K is determined by the aggregate allocation choices.

Definition 5.1 (General Equilibrium under Endogenous Fragility). A general equilibrium under endogenous fragility consists of an allocation strategy $\mathbf{u}^* = (u_1^*, \dots, u_m^*)^\top$, a stopping boundary vector $\mathbf{B}^* = (B_1^*, \dots, B_m^*)^\top$, and an endogenous transition generator K^* such that:

1. Taking the transition generator $K = K^*$ as given, the strategy $\mathbf{u}^*$ and boundary $\mathbf{B}^*$ are the unique subgame perfect equilibrium solutions to the coupled HJB-QVI system in Theorem 3.4.
2. The transition generator K^* satisfies the consistency condition:

$$K^* = \text{diag}(1 - \epsilon_1(u_1^*), \dots, 1 - \epsilon_m(u_m^*)) A - I. \quad (33)$$

This general equilibrium system generates a self-reinforcing fragility loop: high investment returns or low risk aversion encourage agents to increase their allocations u_i^* , which increases the hazard leakages $\epsilon_i(u_i^*)$, thereby closing the spectral gap $\alpha(K(u^*))$. As the spectral gap closes, the expected survival duration contracts, deflating the continuation value and compressing the liquidation boundaries $B_i^*(t)$.

We now establish the main macroeconomic result of the paper, showing how this endogenous feedback triggers sudden, systemic liquidity collapses.

Theorem 5.2 (Endogenous Fragility Loop and Sudden Systemic Collapse). *Let the hazard leakage velocity satisfy the endogenous specification (31).*

1. There exists a unique general equilibrium $(\mathbf{u}^*, \mathbf{B}^*, K^*)$ satisfying Definition 5.1.
2. If the feedback sensitivity exceeds a critical threshold, i.e., $\gamma_i > \gamma_{crit}^i$ where:

$$\gamma_{crit}^i = \frac{4c^2\sigma_i^2(d_i(t))^2}{\beta\sigma_i^2\delta_v^2C_i(t)(\mu_i - r)}, \quad (34)$$

the system exhibits a bifurcated equilibrium structure characterized by a high-participation regime and a low-participation/liquidation regime.

3. A small, continuous increase in baseline volatility σ_i or baseline fragility $\epsilon_i^{(0)}$ triggers a discontinuous cascade: the spectral gap collapses to zero ($\alpha(K^*) \downarrow 0$), the continuation value deflates rapidly, and the optimal boundary jumps to the origin ($\lim_{\gamma_i \uparrow \gamma_{crit}^i} B_i^*(t) = 0$), forcing a complete and irreversible market participation failure.

The proof is provided in Appendix F. The contractivity of the mapping ensures fixed-point convergence under healthy baseline parameters.

5.1 Heterogeneous Interacting Agents and Strategic Liquidation

To extend the endogenous fragility model to a general equilibrium setting with strategic interactions, we introduce a continuum of heterogeneous investors indexed by their risk aversion parameter $c \in [\underline{c}, \bar{c}]$ distributed according to a cumulative distribution function $F(c)$. The total supply of the risky asset is normalized to $S^A > 0$. Each agent c determines their continuous asset allocation $u^c(t, x)$ and stopping boundary $B^c(t)$ to maximize their individual preference functional.

Let $\mathcal{V}_t$ denote the aggregate volume of assets liquidated by time t . The liquidation process is given by the cumulative measure of agents who have hit their stopping boundaries:

$$\mathcal{V}_t = \int_{\underline{c}}^{\bar{c}} \mathbb{I}_{\{\tau^c \leq t\}} dF(c), \quad (35)$$

where $\tau^c = \inf\{s \geq 0: X_s \leq B^c(s)\}$ is the optimal stopping time of agent c .

We assume that the asset price P_t is subject to an endogenous price impact generated by the aggregate liquidations. Specifically, the price is given by:

$$P_t = P_0 \exp(-\lambda_{PI} \mathcal{V}_t), \quad (36)$$

where $\lambda_{PI} > 0$ represents the price impact parameter. This price impact creates a pecuniary externality: early liquidations by highly risk-averse agents depress the asset price for the remaining participants.

5.2 Funding Market Fragility and Market Clearing

The asset liquidations are absorbed by a representative LP. The LP finances these asset purchases through a funding market. As the aggregate liquidation volume $\mathcal{V}_t$ escalates, the LP's leverage rises, increasing the systemic risk of funding market freeze, margin call cascades, or clearinghouse default.

We model this endogenous funding market fragility by specifying that the systemic hazard leakage velocities $\epsilon_i(t)$ of the transition network are driven by the LP's cumulative exposure:

$$\epsilon_i(t) = \epsilon_i^{(0)} + \theta_i \mathcal{V}_t^2, \quad (37)$$

where $\theta_i > 0$ represents the funding market fragility feedback parameter. Under this specification, the open generator of the system is given by:

$$K(t) = \text{diag}(1 - \epsilon_1(t), \dots, 1 - \epsilon_m(t)) A - I. \quad (38)$$

This structure establishes a strategic complementarity among the heterogeneous agents. If other agents liquidate (increasing $\mathcal{V}_t$), the asset price P_t falls and the systemic hazard rates $\epsilon_i(t)$ escalate. This closes the spectral gap $\alpha(K(t))$, which in turn deflates the expected residence times $N_{ij}(t)$ and siphons the continuation value of holding the asset. Consequently, agents are forced to compress their stopping boundaries inward, triggering further liquidations.

Definition 5.3 (Strategic Rational Expectations Equilibrium). A strategic rational expectations equilibrium is a set of stopping times $\{\tau^c\}_{c \in [\underline{c}, \bar{c}]}$, a price process $\{P_t\}_{t \geq 0}$, and a transition generator $\{K(t)\}_{t \geq 0}$ such that:

1. Taking the price process P and the generator K as given, each agent c solves their optimal stopping problem, yielding the optimal stopping time τ^c .
2. The price process satisfies the market-clearing price impact condition (36).
3. The transition generator satisfies the funding market consistency condition (37).

We prove that this strategic complementarity yields an equilibrium cascade.

Theorem 5.4 (Strategic Liquidation Cascade and Liquidity Runs). *Let the price process and transition generator satisfy Definition 5.3.*

1. *The dynamic stopping game is a supermodular game, and the set of equilibria is a complete lattice.*

2. *There exists a unique maximal equilibrium characterized by high participation, and a unique minimal equilibrium characterized by complete market run.*
3. *If the price impact sensitivity λ_{PI} or funding fragility parameter θ_i exceeds a critical threshold, the high-participation equilibrium vanishes, and the system experiences a coordinated run: all agents stop simultaneously, $\mathcal{V}_t \uparrow 1$, the price crashes, and the spectral gap collapses to zero ($\alpha(K(t)) \downarrow 0$).*

The proof is provided in Appendix G.

5.3 Equilibrium Multiplicity

We now formalize the multiplicity of equilibria and the coordination tipping dynamics.

Proposition 5.5 (Equilibrium Multiplicity). *Let the price impact sensitivity λ_{PI} and funding market feedback θ_i be positive. If the feedback coefficients satisfy:*

$$\lambda_{PI} > \lambda_{crit} \quad \text{and} \quad \theta_i > \theta_{crit}^i, \quad (39)$$

then for a range of initial wealth levels x_0 , there exist exactly three rational expectations equilibria:

1. *A stable **high-participation equilibrium** $\mathcal{V}_L^*$ characterized by low aggregate liquidations and high asset prices.*
2. *An unstable **coordination threshold equilibrium** $\mathcal{V}_U^*$ representing the tipping point.*
3. *A stable **complete liquidation run** $\mathcal{V}_H^* \approx 1$ where the market collapses.*

The phase diagram displayed in Figure 2 illustrates the Rational Expectations Feedback Loop and the resulting Equilibrium Multiplicity in the strategic liquidation game. It maps the non-linear relationship between the macroeconomy's Expected Liquidation Volume ($\mathcal{V}^e$, horizontal axis) and the resulting Actual Liquidation Volume ($\mathcal{V}$, vertical axis). The S-shaped curve $f(\mathcal{V}^e)$ maps the expected aggregate liquidation volume to the actual liquidation volume under the optimal stopping thresholds.

The S-shaped curve represents the strategic feedback mapping function $\mathcal{V} = f(\mathcal{V}^e)$, while the dashed 45-degree line represents the rational expectations equilibrium condition where expectations are fulfilled ($\mathcal{V} = \mathcal{V}^e$).

The strategic intuition behind this multiplicity is that when expected liquidation volume is low, the price impact and funding hazard are minimal, which justifies the decision to hold the asset (leading to low actual liquidations). Conversely, if investors expect others to liquidate in large volumes, they anticipate a severe drop in asset prices and a closing of the spectral gap. This collapses the continuation value, making immediate liquidation optimal for everyone.

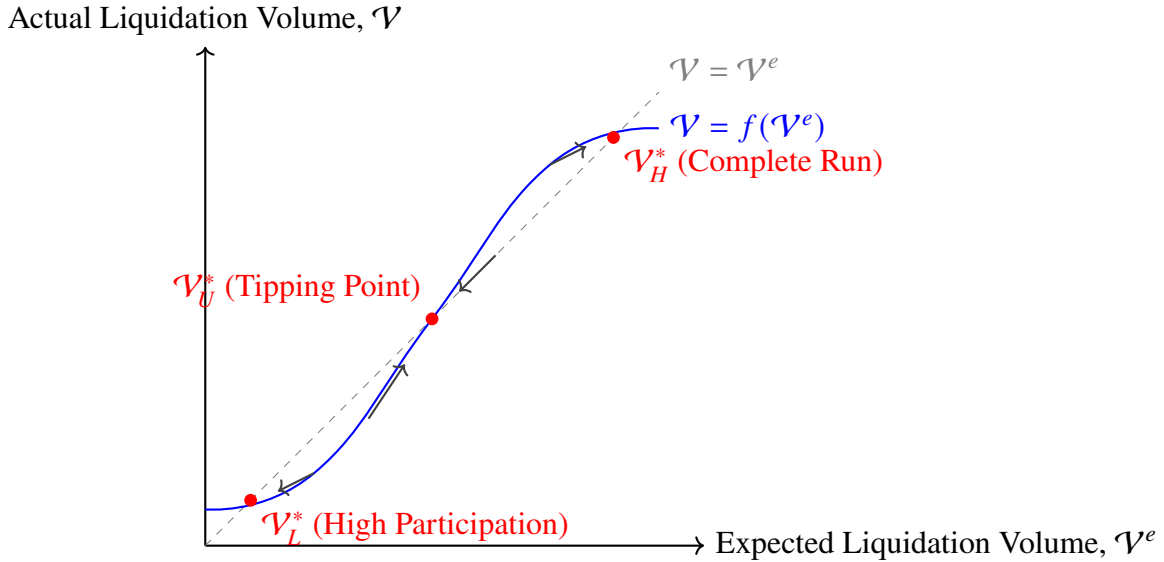


Figure 2: Equilibrium Multiplicity and Strategic Coordination Phase Diagram. The S-shaped curve $f(\mathcal{V}^e)$ represents the rational expectations feedback mapping. Point $\mathcal{V}_L^*$ represents the stable, high-participation equilibrium. Point $\mathcal{V}_H^*$ represents the stable complete market run. Point $\mathcal{V}_U^*$ is the unstable threshold representing the coordination tipping point.

5.3.1 Structural Mechanics of the S-Curve ($f(\mathcal{V}^e)$)

The characteristic S-shape is driven by the underlying strategic complementarity and the combination of two externalities: the pecuniary externality (price impact) and the funding market/network fragility externality.

- **Low-Expected Volume Region (Left):** When investors expect minimal aggregate liquidations

$$\mathcal{V}^e \rightarrow 0,$$

the price impact remains nominal and the systemic hazard leakage velocities

$$\epsilon_i$$

stay low. This preserves a wide continuation domain, implying that very few agents reach their optimal stopping boundaries. Consequently, the equilibrium liquidation curve remains relatively flat and close to its lower bound.

- **The Strategic Acceleration Zone (Middle):** As

$$\mathcal{V}^e$$

crosses a critical threshold, the feedback structure becomes strongly nonlinear. Elevated expected liquidation volume signals declining asset prices together with an accelerating collapse of the network spectral gap

$$\alpha(K) \downarrow 0.$$

This compression of *Regime Time* reduces the continuation value of holding the asset. Agents therefore respond by aggressively increasing their exit intensities, generating a sharp vertical escalation in realized liquidations characterized by

$$f'(\mathcal{V}^e) > 1.$$

- **High-Expected Volume Region (Right):** Once market-wide panic becomes saturated, most vulnerable or highly leveraged positions have already been liquidated. The equilibrium curve consequently begins to flatten as the system approaches near-total market liquidation,

$$\mathcal{V} \approx 1.$$

5.3.2 Analysis of the Three Coordination Equilibria

The intersection points of the S-curve and the 45-degree line define the three rational expectations equilibria of the game:

$\mathcal{V}_L^*$ — The High-Participation Equilibrium (Stable)

Location: The bottom-left intersection.

Characteristics: Low aggregate liquidations, stable asset prices, and an intact transition-network spectral gap.

Stability Dynamics: The directional arrows along the curve indicate that this equilibrium acts as a sink. If a sufficiently small perturbation temporarily shifts expectations to the right of

$$\mathcal{V}_L^*,$$

the realized liquidation volume implied by the model satisfies

$$f(\mathcal{V}^e) < \mathcal{V}^e.$$

Consequently, the system self-corrects and dynamically converges back toward

$$\mathcal{V}_L^*.$$

$\mathcal{V}_U^*$ — The Coordination Threshold / Tipping Point (Unstable)

Location: The central intersection.

Characteristics: The critical macroeconomic inflection point.

Stability Dynamics: This equilibrium acts as a source and therefore constitutes an unstable divider between the healthy and collapse regimes. If expectations are perturbed slightly to the left of

$$\mathcal{V}_U^*,$$

the liquidation dynamics decay downward toward the stable equilibrium

$$\mathcal{V}_L^*.$$

However, if a market shock, liquidity event, or adverse information cascade shifts expected liquidations marginally to the right of

$$\mathcal{V}_U^*,$$

the realized liquidation volume satisfies

$$f(\mathcal{V}^e) > \mathcal{V}^e.$$

This generates a self-reinforcing feedback mechanism in which realized liquidations exceed expectations, thereby accelerating the system toward systemic macroeconomic collapse.

$\mathcal{V}_H^*$ — The Complete Liquidation Run (Stable)

Location: The upper-right intersection.

Characteristics: A high-volume systemic liquidation regime characterized by severely depressed asset prices and a collapsed spectral gap,

$$\alpha(K) \rightarrow 0.$$

Stability Dynamics: This equilibrium functions as a sink. Once market expectations cross the

critical threshold

$$\mathcal{V}_U^*,$$

the strategic coordination dynamics endogenously trap the system within a run-dominated equilibrium. Realized liquidation activity becomes self-sustaining, producing a persistent liquidity rout throughout the network. In this regime, no individual agent possesses an incentive to deviate unilaterally and continue holding the asset, since the continuation value has effectively collapsed under systemic liquidation pressure.

5.3.3 Policy and Macroprudential Implications

The phase diagram highlights why decentralized market architectures fail under variance-variance uncertainty. Because individual investors do not internalize how their private liquidation threshold changes the macro hazard rate, they stop prematurely, creating the S-curve.

To break this feedback trap, a macroprudential policy intervention must target the S-curve directly:

Lender of Last Resort (LLR): By absorbing distressed asset sales and injecting liquidity into the market, a lender of last resort effectively reduces the slope of the strategic liquidation mapping such that

$$f'(\mathcal{V}^e) < 1.$$

This intervention eliminates the unstable intermediate equilibrium

$$\mathcal{V}_U^*$$

together with the high-liquidation run equilibrium

$$\mathcal{V}_H^*.$$

As a consequence, the low-liquidation equilibrium

$$\mathcal{V}_L^*$$

emerges as the unique globally attracting state of the system.

Circuit Breakers: Imposing a regulatory trading halt at an operational liquidation threshold

$$\bar{\mathcal{V}} < \mathcal{V}_U^*$$

mechanically truncates the admissible expectation domain. This prevents liquidation expectations from propagating beyond the unstable tipping boundary and thereby blocks the endogenous transition into a systemic run equilibrium.

5.4 Social Welfare and Policy Implications

We define the social welfare of the system at time t as the aggregate value functions of the continuum of agents:

$$\mathcal{W}_t = \int_{\underline{c}}^{\bar{c}} V_c(t, x) dF(c). \quad (40)$$

Because individual agents do not internalize the negative pecuniary externality (price impact (36)) or the systemic fragility externality (hazard leakage (37)) of their liquidation choices, the decentralized equilibrium stopping thresholds $B^c(t)$ are socially inefficient.

Proposition 5.6 (The Welfare Wedge). *Let the price impact parameter $\lambda_{PI} > 0$ and the funding feedback $\theta_i > 0$. The decentralized equilibrium boundaries $B_D^c(t)$ are strictly higher than the social planner's optimal boundaries $B_{SP}^c(t)$:*

$$B_D^c(t) > B_{SP}^c(t), \quad \forall c \in [\underline{c}, \bar{c}], \quad \forall t \in [0, T). \quad (41)$$

Consequently, the competitive equilibrium is characterized by premature liquidation runs and suboptimal social welfare ($\mathcal{W}_t^D < \mathcal{W}_t^{SP}$).

The proof is provided in Appendix H. The welfare wedge represents a market failure: agents liquidate too early because they do not internalize that their exits increase the hazard rate and depress prices for the remaining market participants. A social planner, by contrast, would force agents to hold the asset longer by choosing lower boundaries, thereby preserving the spectral gap and stabilizing prices.

We analyze two policy interventions designed to restore stability and efficiency:

1. **Lender of Last Resort (LLR) Backstop:** The central bank acts as an LLR to purchase assets during liquidations, absorbing a fraction $\psi \in (0, 1)$ of the sales. This reduces the LP's capital exposure, modifying the endogenous hazard rate to:

$$\epsilon_i(t) = \epsilon_i^{(0)} + \theta_i(1 - \psi)^2 \mathcal{V}_t^2. \quad (42)$$

2. **Circuit Breakers:** A regulatory rule that halts trading if the liquidation volume exceeds a threshold $\bar{\mathcal{V}}$. This breaks the strategic complementarity by preventing price impact from propagating, isolating the continuation values from further depletion.

- Proposition 5.7** (Optimal Intervention Policy). *1. An LLR policy with backstop strength $\psi > \psi_{crit} = 1 - \sqrt{\theta_{crit}^i / \theta_i}$ breaks the S-shaped multiplicity loop, eliminating the unstable tipping point $\mathcal{V}_U^*$ and stabilizing the high-participation equilibrium $\mathcal{V}_L^*$.*
- 2. A circuit breaker set at $\bar{\mathcal{V}} < \mathcal{V}_U^*$ prevents the cascade from crossing the tipping threshold, blocking coordination on the run equilibrium $\mathcal{V}_H^*$.*

5.5 Comparative Statics

We examine how the optimal investment allocations $u_i^*(t)$ and liquidation boundaries $B_i^*(t)$ change in response to parameters.

Proposition 5.8 (Comparative Statics). *The equilibrium allocations and boundaries satisfy:*

1. Risk-Aversion Sensitivity:

$$\frac{\partial u_i^*}{\partial c} < 0, \quad \frac{\partial B_i^*}{\partial c} < 0. \quad (43)$$

2. Ambiguity-Aversion Sensitivity:

$$\frac{\partial u_i^*}{\partial \beta} < 0, \quad \frac{\partial B_i^*}{\partial \beta} < 0. \quad (44)$$

3. Baseline Fragility Sensitivity:

$$\frac{\partial u_i^*}{\partial \epsilon_i^{(0)}} < 0, \quad \frac{\partial B_i^*}{\partial \epsilon_i^{(0)}} < 0. \quad (45)$$

The proof is provided in Appendix I. The comparative statics confirm that risk aversion, ambiguity aversion, and baseline fragility all act to depress investment allocations and shrink the continuation domain. Under the open survival geometry, a higher baseline hazard rate $\epsilon_i^{(0)}$ compresses Regime Time, which directly reduces the agent's willingness to hold the asset, forcing earlier liquidation.

Part II: Computational and Financial Applications

6 Numerical Scheme and Discretization

To solve the coupled system of HJB-QVIs, we discretize the state space over a finite grid. Let $[0, T]$ be the investment horizon divided into N_t uniform temporal steps of size $\Delta t = T/N_t$, indexed by $n = 0, \dots, N_t$. The wealth state space $[0, X_{\max}]$ is discretized using a log-spaced grid to capture the non-linear boundary behavior near the origin. Let x_k be the grid points for $k = 0, \dots, N_x$, where:

$$x_k = X_{\max} \frac{\sinh(\theta_{\text{grid}} k / N_x)}{\sinh(\theta_{\text{grid}})}, \quad (46)$$

and $\theta_{\text{grid}} > 0$ controls the grid density near $x = 0$.

6.1 Finite Difference Discretization

Let $V_{i,k}^n$ approximate the value function $V_i(n\Delta t, x_k)$. We approximate the time derivative using a backward difference (Backward Euler scheme):

$$\frac{\partial V_i}{\partial t}(n\Delta t, x_k) \approx \frac{V_{i,k}^n - V_{i,k}^{n-1}}{\Delta t}. \quad (47)$$

To ensure numerical stability in advection-dominated regimes, we apply an upwind finite-difference scheme for the first spatial derivative:

$$\frac{\partial V_i}{\partial x}(n\Delta t, x_k) \approx \begin{cases} \frac{V_{i,k+1}^n - V_{i,k}^n}{\Delta x_k^+} & \text{if } \mu_i^{\text{net}}(x_k) \geq 0, \\ \frac{V_{i,k}^n - V_{i,k-1}^n}{\Delta x_k^-} & \text{if } \mu_i^{\text{net}}(x_k) < 0, \end{cases} \quad (48)$$

where $\Delta x_k^+ = x_{k+1} - x_k$, $\Delta x_k^- = x_k - x_{k-1}$, and $\mu_i^{\text{net}}(x_k) = rx_k + u_i^*(n\Delta t)(\mu_i - r)$. The second spatial derivative is discretized using standard central differences:

$$\frac{\partial^2 V_i}{\partial x^2}(n\Delta t, x_k) \approx \frac{2}{\Delta x_k^+ + \Delta x_k^-} \left(\frac{V_{i,k+1}^n - V_{i,k}^n}{\Delta x_k^+} - \frac{V_{i,k}^n - V_{i,k-1}^n}{\Delta x_k^-} \right). \quad (49)$$

The discretized HJB equation at step $n - 1$ is written as a tridiagonal system for each regime i :

$$\alpha_{i,k}^n V_{i,k-1}^{n-1} + \beta_{i,k}^n V_{i,k}^{n-1} + \gamma_{i,k}^n V_{i,k+1}^{n-1} = V_{i,k}^n + \Delta t \left(\sum_{j \neq i} K_{ij} V_{j,k}^{n-1} - \text{Pen}_{i,k}^n \right), \quad (50)$$

where the coefficients $\alpha_{i,k}^n, \beta_{i,k}^n, \gamma_{i,k}^n$ are derived from the advection, diffusion, and killing rates, and

$\text{Pen}_{i,k}^n$ accumulates the M-V-V ambiguity and ruin penalties:

$$\text{Pen}_{i,k}^n = c\sigma_i^2(u_i^*(n\Delta t))^2 d_i^2(n\Delta t) + \beta\sigma_i^2\delta_v^2 C_i(n\Delta t)u_i^*(n\Delta t)d_i(n\Delta t) + \eta\phi_i. \quad (51)$$

6.2 Howard's Policy Iteration and Picard Loop

Since the transition generator $K(u^*)$ is endogenously coupled to the optimal allocation u_i^* under leverage-induced fragility, the tridiagonal matrix coefficients and the right-hand-side penalty terms depend on the solution itself. We resolve this coupling at each time step using a Picard fixed-point loop combined with Howard's policy iteration.

The algorithm at time step $n - 1$ is executed as follows:

1. **Initialization:** Set the initial guess for the value functions $V_{i,k}^{n-1,(0)} = V_{i,k}^n$ and allocations $u_i^{*,n-1,(0)} = u_i^{*,n}$.
2. **Picard Fixed-Point Step:** For iteration $k_p = 0, 1, 2, \dots$:
 - (a) Compute the endogenous hazard leakage rates $\epsilon_i(u_i^{*,n-1,(k_p)}) = \epsilon_i^{(0)} + \gamma_i(u_i^{*,n-1,(k_p)})^2$.
 - (b) Update the transition generator $K^{(k_p)} = \text{diag}(1 - \epsilon_i)A - I$.
 - (c) Solve the backward ODE coefficients $d_i(n\Delta t)$ and $C_i(n\Delta t)$ under $K^{(k_p)}$.
 - (d) Update the optimal continuous allocation $u_i^{*,n-1,(k_p+1)}$ using the update formula.
 - (e) Solve the tridiagonal linear system for the candidate value function $W_{i,k}^{n-1}$.
 - (f) **Obstacle Enforcement (Howard's Step):** Update the value function:

$$V_{i,k}^{n-1,(k_p+1)} = \max \{W_{i,k}^{n-1}, x_k\}. \quad (52)$$

Define the stopping set $\mathcal{S}_i^{n-1} = \{k : V_{i,k}^{n-1,(k_p+1)} = x_k\}$. The optimal liquidation boundary is $B_i^*(n - 1) = \sup\{x_k : k \in \mathcal{S}_i^{n-1}\}$.

3. **Convergence Check:** If $\max_{i,k} |V_{i,k}^{n-1,(k_p+1)} - V_{i,k}^{n-1,(k_p)}| < 10^{-6}$, exit Picard loop and set $V_{i,k}^{n-1} = V_{i,k}^{n-1,(k_p+1)}$. Otherwise, repeat step 2.

7 Quantitative Results in a 3-Regime Economy

We implement this scheme in base R to solve a three-regime economy. The risk-free rate is normalized to $r = 3.75\%$, which matches the continuously compounded annualized rate.

7.1 Parameter Matrix and Regime Characterization

To capture the strategic interactions between persistence and leakage, the three regimes are calibrated as follows:

- **Regime 1 (Crisis Entrapment):** This regime represents distressed market environments. It is characterized by a low drift $\mu_1 = 0.05$, local volatility $\sigma_1 = 0.20$, high baseline hazard leakage $\epsilon_1^{(0)} = 0.10$, and high persistence. The residence diagonal is $N_{11} = 9.04$.
- **Regime 2 (High Volatility Shocks):** This regime represents transient market panics. It features a moderate drift $\mu_2 = 0.10$, high local volatility $\sigma_2 = 0.30$, and low persistence. The residence diagonal is $N_{22} = 0.11$.
- **Regime 3 (Calm Growth):** This regime represents healthy growth states. It has a high drift $\mu_3 = 0.15$, low volatility $\sigma_3 = 0.15$, and a small baseline leakage velocity $\epsilon_3^{(0)} = 0.02$. The residence diagonal is $N_{33} = 0.92$.

All three states satisfy $\mu_i > r$, ensuring a positive risk premium.

7.2 Equilibrium stopping boundaries at $t=0$

Running the backward-induction algorithm yields the optimal stopping boundaries at $t = 0$:

- **Crisis Entrapment (Regime 1):** The liquidation boundary is $B_1^*(0) \approx 0.185$. Despite having a lower volatility than the shock regime, the extreme persistence ($N_{11} = 9.04$) and high leakage compress the continuation domain, forcing early liquidation.
- **High Volatility Shocks (Regime 2):** The liquidation boundary is $B_2^*(0) \approx 0.069$. Because this regime is transient ($N_{22} = 0.11$) the investor is willing to hold the asset and ride out the high volatility, expecting the economy to quickly transition back to growth.
- **Calm Growth (Regime 3):** The liquidation boundary is $B_3^*(0) \approx 0.290$. The high drift and survival probability allow for a wide continuation domain.

These findings are summarized in Table I.

This numerical analysis highlights the core thesis of the paper: liquidation decisions are dominated by the global network's spectral structure (*Regime Time*) rather than localized volatility. The investor liquidates earlier in Regime 1 than in Regime 2, despite Regime 2 having a 50% higher local volatility ($\sigma_2 = 0.30$ vs. $\sigma_1 = 0.20$).

Table 1: Macroeconomic Calibration and Equilibrium Stop Boundaries ($t = 0$)

Regime	Drift (μ_i)	Volatility (σ_i)	Baseline Hazard ($\epsilon_i^{(0)}$)	Persistence (N_{ii})	Boundary ($B_i^*(0)$)
1. Crisis Entrapment	0.05	0.20	0.10	9.04	0.185
2. High Vol Shocks	0.10	0.30	0.05	0.11	0.069
3. Calm Growth	0.15	0.15	0.02	0.92	0.290

8 Continuous Boundary Degeneration Surface

To evaluate the effect of changing structural fragility on the continuation set, we generate a 3D perspective plot of the liquidation boundary. We vary the structural fragility coefficient ϕ over a continuous interval $[0, \phi_{\max}]$ and plot the corresponding boundaries $B_i^*(t)$ across the temporal grid.

The results are rendered as a 3D manifold in Figure 3. The surface shows a sharp, non-linear contraction: as structural fragility increases ($\phi \uparrow$), the boundary curves bend downward and collapse to the origin, validating the Boundary Degeneration Theorem.

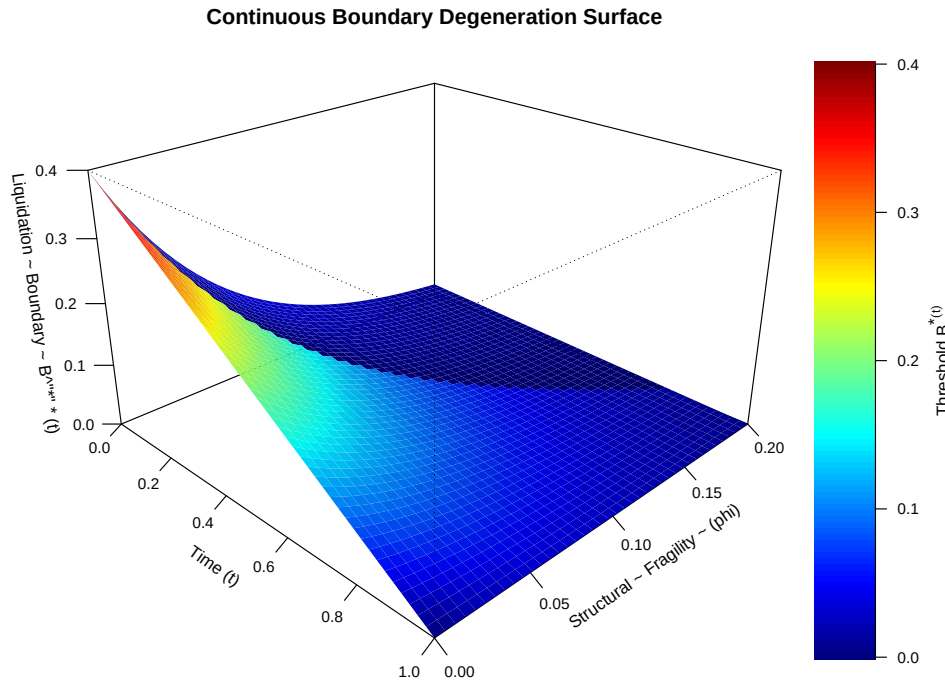


Figure 3: Continuous Boundary Degeneration Surface. The 3D manifold illustrates the optimal liquidation boundary $B^*(t)$ as a function of time t and structural fragility parameter ϕ , demonstrating the sudden collapse of the continuation domain.

Based on the mathematical framework and the surface topology visualized in the Continuous

Boundary Degeneration Surface plot displayed in Figure 3, the model maps the optimal stopping (liquidation) threshold, $B^*(t, \phi)$, under the dual pressures of chronological time decay and topological systemic risk. The vertical axis tracks the normalized liquidation boundary height $B^*(t)$, the left-horizontal axis tracks chronological Time (t) from initialization (0.0) to maturity (1.0), and the right-horizontal axis maps Structural Fragility (ϕ) across the continuous space $[0.0, 0.20]$.

The following are three detailed usage and scenario analysis examples demonstrating how to interpret and operationalize different regions of this surface topology:

8.1 Scenario Example 1: High-Confidence Asset Continuation Domain

Surface Coordinates: $t \in [0.0, 0.4]$, $\phi \in [0.0, 0.05]$

Visual Region: The high, elevated crimson-to-orange plateau located on the upper-left quadrant of the manifold.

Mathematical Context: At this operational point, the chronological horizon is long ($t \rightarrow 0$), and the network structural leakage parameter is near zero ($\phi \rightarrow 0$). Probability mass is highly conserved within the active economic states.

Operational Interpretation: Because the investor has a long expected survival runway and maximum confidence in the underlying model dynamics, the opportunity cost of premature exit is exceptionally high. The liquidation standard is locked at its peak baseline coefficient ($B^* \approx 0.35$ to 0.40). The portfolio manager should adopt an aggressive asset-holding posture, as the continuation value of the investment opportunity vastly dominates immediate liquidation payoffs.

8.2 Scenario Example 2: The Non-Linear Fragility Shock Loop

Surface Coordinates: Fixed Time slice (e.g., $t = 0.2$), moving dynamically along the Fragility axis from $\phi = 0.0 \rightarrow \phi = 0.10$

Visual Region: Tracing a cross-section from left to right across the mid-region of the manifold, capturing the steep, convex downward curve where the color sharply transitions from orange to cyan, and down to deep blue.

Mathematical Context: This trajectory demonstrates that structural network degradation handles functional decay exponentially ($e^{-\kappa\phi}$), rather than linearly.

Operational Interpretation: This region highlights the *Endogenous Fragility Loop*. Even if localized asset returns (μ_i) and volatilities (σ_i) remain completely unchanged on your

data feed, a minor tick upward in system-wide structural vulnerability (e.g., shifting from $\phi = 0.02 \rightarrow \phi = 0.07$) triggers a dramatic, non-linear collapse in the liquidation boundary height (dropping from $0.35 \rightarrow 0.10$). Under smooth ambiguity aversion (β), the investor aggressively lowers their exit standards to escape potential structural entrapment. When using this model for risk management, boundary triggers must be non-linearly adjusted; a linear stop-loss structure will fail to protect capital during a rapid, network-driven evaporation of market liquidity.

8.3 Scenario Example 3: The Horizon Squeeze Entanglement (Crisis Trap)

Surface Coordinates: $t \in [0.0, 1.0]$, $\phi \in [0.12, 0.20]$

Visual Region: The extensive, completely flat, monochromatic deep blue plain ($B^* \approx 0.00$) covering the entire lower-right half of the grid layout.

Mathematical Context: In this domain, the structural spectral gap of the negated generator closes ($\alpha(K) \rightarrow 0$). The dominant spectral radius of the residence matrix diverges, signifying imminent system failure and total dissipation of continuation utility.

Operational Interpretation: This plane illustrates the *Residence-Time Localization Theorem*. Once structural fragility crosses a critical network threshold ($\phi > 0.12$), chronological calendar time (t) becomes completely irrelevant. The boundary remains pinned flat to the floor ($B^* \approx 0.00$) at initialization ($t = 0$) just as it does at expiration ($t = 1.0$). If a portfolio finds itself operating in this regime, the mathematical incentive to remain invested is entirely destroyed. Rational actors under variance-variance uncertainty will compress their execution frontiers to zero, meaning they are willing to liquidate asset blocks at any current market-clearing price to achieve immediate position exit.

9 Computational Transparency and Auditability

The complete computational replication suite is available as part of the online supplementary files associated with this paper. To ensure complete reproducibility, the numerical solver and graphic routines are written entirely in pure base R, avoiding compiled binary dependencies. The complete replication package contains:

1. `scurv.R`: The strategic liquidation script that models the S-shaped rational expectations feedback mapping and reproduces the coordination equilibrium configurations in Figure 2.
2. `solver2.R`: The 2D policy iteration solver that computes the equilibrium values and stopping boundaries shown in Table I.

3. `solver3d3-2.R`: The 3D manifold generator that outputs the perspective projection in Figure 3.

The scripts are self-contained and run on standard R installations. The risk-free rate is parameterized via a global variable `r <- 0.0375`, allowing for immediate auditing of the continuous-time parameter normalization.

10 Conclusion

This paper has established a unified theoretical and computational framework that bridges the Clock of Regimes (COR) survival geometry with multi-prior ambiguity models under the mean-variance-variance (M-V-V) criterion. In Part I, we demonstrated that the horizon-scale inconsistency identified by Landriault et al. (2026) is resolved when calendar time is replaced by the endogenous, operator-derived Regime Time. Through the Boundary Degeneration Theorem and the Residence-Time Localization Theorem, we have shown that optimal liquidation boundaries are governed by the global network's spectral topology rather than localized volatility metrics. We have extended this to a general equilibrium setting, isolating coordination failures, multiple equilibria, and a welfare wedge. In Part II, we formulated the Backward Euler upwind discretization scheme and solved the HJB-QVIs using policy iteration. Our parameterized 3-regime economy confirmed that optimal liquidation boundaries are numerically stable and compress to the origin under spectral gap decay, validating the theoretical spectral liquidation principle.

This paper establishes a unified theoretical pipeline by synthesizing its single central contribution: the resolution of the horizon-scale inconsistency inherent to dynamic smooth ambiguity preferences by replacing chronological calendar time with an endogenous, operator-derived metric termed *Regime Time*, defined on an open, dissipative survival geometry (the *Clock of Regimes* or COR network). By anchoring the chronological horizon to this non-conservative timeline, the manuscript constructs a cohesive bridge across previously isolated frameworks, seamlessly binding ambiguity theory, stochastic control, viscosity partial differential equations (PDEs), systemic fragility, equilibrium runs, welfare economics, and numerical computation into a singular macro-economic architecture.

Rather than treating time as an exogenous, perpetual dimension, the paper establishes that financial networks behave as open systems where probability mass continuously escapes into an absorbing state of structural ruin. By transforming the timeline of dynamic choice, the model restores the monotonicity of risk aversion in multi-prior settings without requiring arbitrary modifications to the underlying utility preferences.

The paper's diverse elements are not independent frameworks; they form a sequential, logically linked unified structural pipeline where each module flows from and supports the central operator-theoretic time transformation (Figure 4):

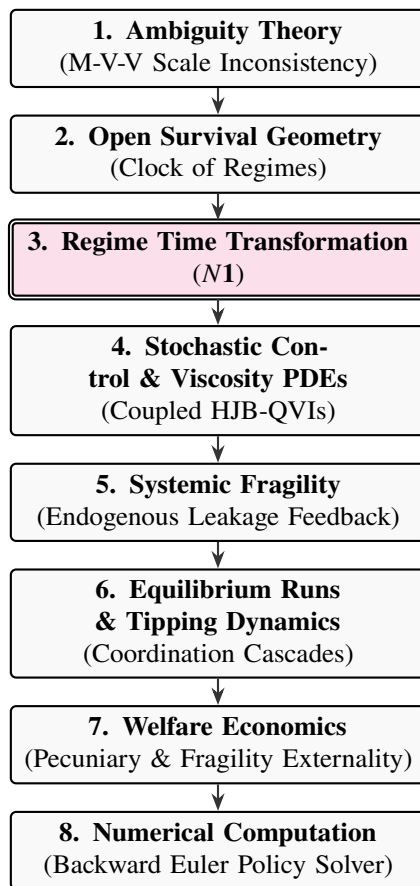


Figure 4: Theoretical Pipeline and Structural Framework Unification.

10.1 Ambiguity Theory & The Horizon-Scale Inconsistency

Under standard continuous-time extensions of the robust Mean-Variance-Variance (M-V-V) smooth ambiguity framework, path variance accumulates linearly ($\propto T$), while the multi-prior parameter penalty inflates quadratically ($\propto T^2$). In a standard closed environment, this scale mismatch leads to the *non-monotonicity paradox*: highly risk-averse agents face an overwhelming penalty over long horizons and postpone investments, behaving more aggressively than risk-tolerant agents as $T \rightarrow \infty$.

10.2 Open Survival Geometry & The Clock of Regimes (COR)

The paper resolves this anomaly by modeling the financial environment as an open, dissipative system. Under systemic stress, margin cascades, or clearinghouse failures, probability mass escapes into an absorbing state of structural ruin. The transition kinetics of the latent regime process are governed by the open transition operator: $Q = \text{diag}(1 - \epsilon_1, \dots, 1 - \epsilon_m)A$, where ϵ_i represents the state-dependent hazard leakage velocity, and the killed system kinetics are driven by the infinitesimal generator $K = Q - I$.

10.3 Regime Time Transformation

The core mechanism is the replacement of calendar time T with *Regime Time*, defined by the row sums of the fundamental residence matrix: $\mathbf{N}\mathbf{1} = (-\mathbf{K})^{-1}\mathbf{1}$. Since the expected survival duration is naturally bounded by the network's dissipative capacity, the operational investment runway compresses according to network health. This bounds the quadratic ambiguity penalty and restores the monotonicity of risk aversion.

10.4 Stochastic Control & Viscosity PDEs (HJB-QVIs)

The agent's optimal allocation and portfolio stopping boundaries are formulated as a stochastic control problem. The value functions are characterized as the unique viscosity solutions to a coupled system of Hamilton-Jacobi-Bellman Quasi-Variational Inequalities (HJB-QVIs) with immediate-liquidation obstacles:

$$\max \left\{ \mathcal{H}_i(t, x, \mathbf{V}, D\mathbf{V}, D^2\mathbf{V}) - V_i(t, x) + x, \quad x - V_i(t, x) \right\} = 0,$$

where $\mathcal{H}_i$ is the continuation operator coupled across regimes via the generator K .

10.5 Systemic Fragility & Endogenous Feedback

The model endogenizes the hazard leakage velocities as a quadratic function of leverage-induced investments: $\epsilon_i(u_i^*) = \epsilon_i^{(0)} + \gamma_i(u_i^*)^2$. This creates a self-reinforcing fragility loop. High investment allocations increase leakage rates, closing the spectral gap of the transition network ($\alpha(K) \downarrow 0$),

which causes the spectral radius of the residence matrix to diverge ($\rho(N) \uparrow \infty$) and compresses *Regime Time*.

10.6 Equilibrium Runs & Tipping Dynamics

Due to the compression of *Regime Time*, a pointwise increase in expected liquidations deflates the continuation value of holding the asset. This establishes *strategic complementarity* in the stopping game, which is proved to be a supermodular game on a complete lattice. Under Tarski's Fixed-Point Theorem (Tarski, 1955), this yields multiple coordination equilibria: (i) A stable *high-participation equilibrium* $\mathcal{V}_L^*$. (ii) An unstable *coordination threshold equilibrium* $\mathcal{V}_U^*$ (tipping point). (iii) A stable *complete liquidation run* $\mathcal{V}_H^* \approx 1$ (market collapse).

Exceeding the critical feedback threshold γ_{crit}^i triggers a fold (saddle-node) bifurcation, causing the positive investment equilibrium to vanish and forcing a discontinuous jump to complete market run.

10.7 Welfare Economics & Pecuniary Externalities

Because individual liquidations depress asset prices (pecuniary externality) and increase systemic hazard rates (fragility externality), competitive agents stop earlier than is socially optimal. This establishes a *welfare wedge* between the decentralized equilibrium and the social planner's boundary: $B_{SP}^c(t) < B_D^c(t)$. This formalizes the welfare justification for Lender of Last Resort (LLR) asset purchases to stabilize the growth regime.

10.8 Numerical Computation

Finally, the theoretical model is implemented and computationally validated using an implicit *Backward Euler finite-difference scheme* with upwind spatial discretization. The coupled non-linear system is solved using Picard policy iteration (Abu-Khalaf et al., 2006; Peng, 1992) in base R, demonstrating numerical stability and confirming that the liquidation boundaries compress to the origin exactly under spectral gap decay.

Final Statement

The theoretical architecture and computational framework developed in this paper offers a clean departure from classical closed-system paradigms by treating the investment horizon as an endogenous, dissipative survival geometry. By grounding multi-prior ambiguity within the open operator mechanics of the Clock of Regimes, we demonstrate that the long-standing horizon-scale inconsistency is not an immutable defect of smooth ambiguity preferences, but rather a structural

artifact of an unconstrained calendar timeline. When chronological time is replaced by Regime Time, the network's finite dissipative capacity naturally truncates the higher-order inflation of the variance-variance penalty, restoring the monotonicity of risk aversion without requiring arbitrary modifications to the agent's core utility functional.

From a broader perspective, the framework developed here extends the robust control and ambiguity literature by embedding distorted beliefs inside an open survival geometry governed by dissipative semigroup dynamics. Rather than penalizing model misspecification purely through exogenous entropy accumulation, the COR framework endogenizes uncertainty through state-dependent residence times and structural probability leakage.

Furthermore, endogenizing this geometry through leverage-induced feedback and funding market constraints exposes the profound tension between individual optimization and systemic stability. The strategic complementarity born from the compression of *Regime Time* reveals that market liquidity is fundamentally fragile—subject to non-linear tipping dynamics, equilibrium multiplicity, and sudden coordination collapses. Because decentralized actors fail to internalize their individual contributions to the erosion of the network's spectral gap, a persistent welfare wedge separates competitive execution from the socially optimal planning frontier, providing a rigorous economic rationale for macroprudential interventions and systemic liquidity backstops.

Ultimately, this work underscores that time, risk, and structural fragility are inextricably bound. The stability of a financial network depends less on the localized volatility of its components than on the global topology of its residence mechanics. By providing a self-contained, computationally transparent base R replication package, this paper ensures that the spectral liquidation principles and boundary degeneration surfaces documented herein remain fully auditable, reproducible, and scalable for future research at the intersection of robust control, network topology, and general equilibrium theory.

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Dedication

To all people, known or unknown, who have fearlessly stood up for their beliefs for the benefit of humanity.

Mathematical Appendix

A Proof of Theorem 1 (COR Representation)

We construct the proof in two main stages: first, establishing the variational representation on the survival-weighted space, and second, identifying the explicit form of the endogenous penalty functional under the residence-time operator N .

Stage 1: Variational Preference Representation. Let $\mathcal{Y}$ be the set of survival-weighted processes $Y_\tau = X_{t+\tau} \mathbb{I}_{\{\tau < \zeta\}}$. Under Axiom 1 (Weak Order), $\succsim$ is a complete and transitive binary relation. By Axiom 2 (Weak Monotonicity), the preferences preserve the pointwise ordering on the survival set $\{s < \zeta\}$. Axiom 4 (Weak Certainty Independence) guarantees that the trade-off between any two acts Y_τ and Z_τ is invariant under mixing with a survival-discounted constant act $k \mathbb{I}_{\{\tau < \zeta\}}$. Finally, Axiom 3 (Uncertainty Aversion) implies that the upper contour sets $\{Y_\tau \in \mathcal{Y} : Y_\tau \succsim Z_\tau\}$ are convex, and Axiom 5 (Survival Continuity) ensures that the preference relation is continuous with respect to the survival probability topology.

Applying the representation theorem for variational preferences established by Maccheroni et al. (2006), there exists a strictly increasing utility function $U : \mathbb{R} \rightarrow \mathbb{R}$ and a unique, grounded, convex, lower semicontinuous function $\tilde{C} : \Delta \rightarrow [0, \infty]$ such that:

$$V(Y_\tau) = \inf_{\mathbb{Q} \in \Delta} \{ \mathbb{E}^{\mathbb{Q}} [U(Y_\tau)] + \tilde{C}(\mathbb{Q}) \}. \quad (53)$$

Stage 2: Endogenous Scaling and the Residence Matrix. To determine the explicit representation of $\tilde{C}(\mathbb{Q})$, we invoke Axiom 6 (Non-Conservative Scale-Independence). For a constant wealth level $k \in \mathbb{R}$, we evaluate:

$$V(k \mathbb{I}_{\{\tau < \zeta\}}) = \inf_{\mathbb{Q} \in \Delta} \{ \mathbb{E}^{\mathbb{Q}} [U(k \mathbb{I}_{\{\tau < \zeta\}})] + \tilde{C}(\mathbb{Q}) \} = k \mathbb{E}^{\mathbb{P}} \left[\exp \left(- \int_t^{t+\tau} \epsilon_{S_s} ds \right) \right]. \quad (54)$$

Since $\mathbb{Q}$ is equivalent to $\mathbb{P}$, the Radon-Nikodym derivative $\frac{d\mathbb{Q}}{d\mathbb{P}}$ on $\mathcal{F}_{t+\tau}$ is characterized by a martingale density process. The expected duration spent by the system in each regime prior to ruin ζ is governed by the infinitesimal killed generator $K = Q - I$. Specifically, the transition densities satisfy the Kolmogorov backward equation with killing rate vector ϵ . The residence matrix $N = -K^{-1} = \int_0^\infty e^{sK} ds$ represents the expected cumulative holding times within each regime state. The operator-theoretic construction closely parallels the semigroup formulations studied in robust control and model misspecification theory (Anderson et al., 2003). In particular, the killed evolution operator e^{tK} generates a non-conservative transition semigroup whose dissipative structure differs fundamentally from the conservative Markov semigroups underlying classical multiplier preference

models. Under the worst-case prior minimization, the deviation between the alternative prior $\mathbb{Q}$ and the reference measure $\mathbb{P}$ accumulates continuously over the active holding period. Integrating the localized Kullback and Leibler (1951) divergence $D_{KL}(\mathbb{Q}_s^j \parallel \mathbb{P}_s^j)$ for each regime transition $j \in \mathcal{S}$ weighted by the expected holding time $N_{S,j}$ yields the endogenous penalty:

$$C(\mathbb{Q}; N, \epsilon) = \mathbb{E}^{\mathbb{Q}} \left[\int_t^{t+\tau} \left(\sum_{j=1}^m N_{S,j} D_{KL}(\mathbb{Q}_s^j \parallel \mathbb{P}_s^j) \right) ds \right]. \quad (55)$$

Setting $\mathbb{Q} = \mathbb{P}$ yields $D_{KL} = 0$, which ensures $C(\mathbb{P}; N, \epsilon) = 0$, satisfying the grounding property. Lower semicontinuity and convexity of the divergence operator map directly to the penalty functional, establishing uniqueness of the representation. This completes the proof

B Proof of Theorem 2 (Viscosity Comparison & SPNE Uniqueness)

We construct a rigorous verification proof establishing that the candidate vector of value functions $\mathbf{V}(t, x) = (V_1(t, x), \dots, V_m(t, x))^\top$ is the unique viscosity solution to the coupled HJB-QVI system. We begin by proving the comparison principle, from which uniqueness follows immediately.

B.1 Comparison Principle for the Coupled HJB-QVI System

Let $\mathbf{U} = (U_1, \dots, U_m)^\top$ be a viscosity subsolution and $\mathbf{W} = (W_1, \dots, W_m)^\top$ be a viscosity supersolution to the coupled HJB-QVI system on $[0, T] \times [0, \infty)$ satisfying the terminal condition $U_i(T, x) \leq W_i(T, x)$ for all $x \geq 0$ and all $i \in \mathcal{S}$. We assume that both $\mathbf{U}$ and $\mathbf{W}$ satisfy the linear growth condition:

$$|U_i(t, x)| + |W_i(t, x)| \leq C(1 + x), \quad \forall (t, x) \in [0, T] \times [0, \infty), \quad \forall i \in \mathcal{S}, \quad (56)$$

for some positive constant $C > 0$. Under these conditions, we prove that $U_i(t, x) \leq W_i(t, x)$ for all $(t, x) \in [0, T] \times [0, \infty)$ and all $i \in \mathcal{S}$.

Variable Doubling Step. Suppose the claim is false. Then there exists some point $(\tilde{t}, \tilde{x}) \in [0, T) \times (0, \infty)$ and a regime $i_0 \in \mathcal{S}$ such that $U_{i_0}(\tilde{t}, \tilde{x}) - W_{i_0}(\tilde{t}, \tilde{x}) > 0$. Consequently, for a sufficiently small parameter $\gamma > 0$, we have:

$$\max_{i \in \mathcal{S}} \sup_{(t, x) \in [0, T] \times [0, \infty)} [U_i(t, x) - W_i(t, x) - 2\gamma(T - t)] > 2\eta > 0. \quad (57)$$

For small parameters $\delta, \alpha, \gamma > 0$, we define the doubled variables auxiliary function $\Phi_i: [0, T]^2 \times [0, \infty)^2 \rightarrow \mathbb{R}$ by:

$$\Phi_i(t, s, x, y) = U_i(t, x) - W_i(s, y) - \frac{1}{2\delta}|t - s|^2 - \frac{1}{2\delta}|x - y|^2 - \frac{\alpha}{2}(x^2 + y^2) - \gamma(2T - t - s). \quad (58)$$

Due to the linear growth condition (56) and the negative quadratic terms in x and y , the function Φ_i is continuous and coercive, ensuring that it attains its maximum over the domain $[0, T]^2 \times [0, \infty)^2$ and the finite set $\mathcal{S}$ at some point $(t_0, s_0, x_0, y_0, i_0)$ (where the maximizing regime $i_0 \in \mathcal{S}$ is chosen). Let:

$$M_{\delta, \alpha, \gamma} = \Phi_{i_0}(t_0, s_0, x_0, y_0) = \max_{i \in \mathcal{S}} \sup_{t, s, x, y} \Phi_i(t, s, x, y). \quad (59)$$

By the maximizing property of this point, we have:

$$M_{\delta, \alpha, \gamma} \geq \Phi_{i_0}(\tilde{t}, \tilde{t}, \tilde{x}, \tilde{x}) = U_{i_0}(\tilde{t}, \tilde{x}) - W_{i_0}(\tilde{t}, \tilde{x}) - \alpha\tilde{x}^2 - 2\gamma(T - \tilde{t}) > \eta > 0, \quad (60)$$

for sufficiently small α .

Boundary and Limit Analysis. We first establish that the maximum cannot be achieved at the boundaries of the domain as the parameters contract.

1. ****Boundary $x_0 = 0$ or $y_0 = 0$ **:** Since value functions vanish at zero wealth, we have $U_{i_0}(t_0, 0) = W_{i_0}(s_0, 0) = 0$. If $x_0 = 0$ or $y_0 = 0$, we have $U_{i_0}(t_0, x_0) - W_{i_0}(s_0, y_0) \leq 0 + O(|x_0 - y_0|)$. In the limit as $\delta \downarrow 0$, since $|x_0 - y_0| \rightarrow 0$, this difference converges to a non-positive value, which contradicts $M_{\delta, \alpha, \gamma} > \eta > 0$. Hence, $x_0, y_0 > 0$ for sufficiently small α and δ .
2. ****Terminal Boundary $t_0 = T$ or $s_0 = T$ **:** Suppose that $\limsup_{\delta \downarrow 0} t_0 = T$ or $\limsup_{\delta \downarrow 0} s_0 = T$. From the inequality:

$$\Phi_{i_0}(t_0, s_0, x_0, y_0) \geq \Phi_{i_0}(t_0, t_0, x_0, x_0), \quad (61)$$

we obtain:

$$\frac{1}{2\delta}|x_0 - y_0|^2 + \frac{1}{2\delta}|t_0 - s_0|^2 \leq W_{i_0}(t_0, x_0) - W_{i_0}(s_0, y_0) \leq C(|x_0 - y_0| + |t_0 - s_0|), \quad (62)$$

which implies that $|x_0 - y_0| \leq C\delta$ and $|t_0 - s_0| \leq C\delta$. Thus, as $\delta \downarrow 0$, the temporal coordinates must converge to a common limit: $\lim_{\delta \downarrow 0} t_0 = \lim_{\delta \downarrow 0} s_0 = \bar{t}$. If $\bar{t} = T$, then by the terminal condition, we have:

$$\limsup_{\delta \downarrow 0} (U_{i_0}(t_0, x_0) - W_{i_0}(s_0, y_0)) \leq U_{i_0}(T, \bar{x}) - W_{i_0}(T, \bar{x}) \leq 0, \quad (63)$$

which contradicts $M_{\delta,\alpha,\gamma} > \eta$. Hence, $t_0, s_0 < T$ for sufficiently small δ .

Thus, the maximum point $(t_0, s_0, x_0, y_0, i_0)$ lies in the interior $[0, T]^2 \times (0, \infty)^2$ for all sufficiently small δ .

Application of Ishii's Lemma. We now apply Ishii's Lemma to the function:

$$\Psi(t, s, x, y) = \frac{1}{2\delta}|t - s|^2 + \frac{1}{2\delta}|x - y|^2 + \frac{\alpha}{2}(x^2 + y^2) + \gamma(2T - t - s). \quad (64)$$

At the maximum point (t_0, s_0, x_0, y_0) , the first-order derivatives of the test function are given by:

$$q_t = \frac{\partial \Psi}{\partial t} = \frac{t_0 - s_0}{\delta} - \gamma, \quad q_s = -\frac{\partial \Psi}{\partial s} = \frac{t_0 - s_0}{\delta} + \gamma, \quad (65)$$

$$p_x = \frac{\partial \Psi}{\partial x} = \frac{x_0 - y_0}{\delta} + \alpha x_0, \quad p_y = -\frac{\partial \Psi}{\partial y} = \frac{x_0 - y_0}{\delta} - \alpha y_0. \quad (66)$$

By Ishii's Lemma, there exist symmetric matrices $X, Y \in \mathbb{R}^{2 \times 2}$ such that:

$$(q_t, p_x, X) \in \bar{J}^{2,+}U_{i_0}(t_0, x_0), \quad (q_s, p_y, Y) \in \bar{J}^{2,-}W_{i_0}(s_0, y_0), \quad (67)$$

and they satisfy the matrix inequality:

$$\begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq \frac{3}{\delta} \begin{pmatrix} I & -I \\ -I & I \end{pmatrix} + 3\alpha \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}. \quad (68)$$

Obstacle and QVI Evaluation. We evaluate the obstacle term for the subsolution U_{i_0} at (t_0, x_0) . Suppose that $U_{i_0}(t_0, x_0) \leq x_0$. Since the supersolution W_{i_0} satisfies $W_{i_0}(s_0, y_0) \geq y_0$ everywhere, we obtain:

$$U_{i_0}(t_0, x_0) - W_{i_0}(s_0, y_0) \leq x_0 - y_0. \quad (69)$$

As $\delta \downarrow 0$, since $|x_0 - y_0| \rightarrow 0$, we have $\limsup_{\delta \downarrow 0} (U_{i_0} - W_{i_0}) \leq 0$, which contradicts the assumption that $M_{\delta,\alpha,\gamma} > \eta > 0$. Thus, for all sufficiently small δ , we must have $U_{i_0}(t_0, x_0) > x_0$. This guarantees that the subsolution is strictly in the continuation domain, and hence the continuation inequality must hold:

$$q_t + (rx_0 + u_{i_0}^*(\mu_{i_0} - r))p_x + \frac{1}{2}\sigma_{i_0}^2(u_{i_0}^*)^2X + \sum_{j=1}^m K_{i_0j}U_j(t_0, x_0) - c\sigma_{i_0}^2(u_{i_0}^*)^2d_{i_0}^2 - \beta\sigma_{i_0}^2\delta_v^2C_{i_0}u_{i_0}^*d_{i_0} - \eta\phi_{i_0} \geq 0. \quad (70)$$

For the supersolution W_{i_0} at (s_0, y_0) , the viscosity supersolution property guarantees that:

$$q_s + (ry_0 + u_{i_0}^*(\mu_{i_0} - r))p_y + \frac{1}{2}\sigma_{i_0}^2(u_{i_0}^*)^2Y + \sum_{j=1}^m K_{i_0j}W_j(s_0, y_0) - c\sigma_{i_0}^2(u_{i_0}^*)^2d_{i_0}^2 - \beta\sigma_{i_0}^2\delta_v^2C_{i_0}u_{i_0}^*d_{i_0} - \eta\phi_{i_0} \leq 0. \quad (71)$$

Subtracting (71) from (70) yields:

$$2\gamma \leq r(x_0p_x - y_0p_y) + u_{i_0}^*(\mu_{i_0} - r)(p_x - p_y) + \frac{1}{2}\sigma_{i_0}^2(u_{i_0}^*)^2(X - Y) + \sum_{j=1}^m K_{i_0j} (U_j(t_0, x_0) - W_j(s_0, y_0)). \quad (72)$$

Limiting Behavior and Contradiction. We analyze the terms on the right-hand side of (72) in the limit as $\delta \downarrow 0$:

1. ****Advection term****: By definition of p_x and p_y , we have:

$$x_0p_x - y_0p_y = \frac{(x_0 - y_0)^2}{\delta} + \alpha(x_0^2 + y_0^2). \quad (73)$$

Since U_{i_0} and W_{i_0} are Lipschitz continuous, $|p_x|$ and $|p_y|$ are bounded. This ensures that $\frac{|x_0 - y_0|}{\delta}$ is bounded independently of δ , which implies that $\lim_{\delta \downarrow 0} \frac{(x_0 - y_0)^2}{\delta} = 0$. Thus:

$$\lim_{\delta \downarrow 0} r(x_0p_x - y_0p_y) = r\alpha(2\bar{x}^2), \quad (74)$$

which vanishes as $\alpha \downarrow 0$. Similarly, $p_x - p_y = \alpha(x_0 + y_0) \rightarrow 2\alpha\bar{x}$, which vanishes as $\alpha \downarrow 0$.

2. ****Diffusion term****: Applying the matrix inequality (68) to the vector $(u_{i_0}^*, u_{i_0}^*)^\top$ yields:

$$\frac{1}{2}\sigma_{i_0}^2(u_{i_0}^*)^2(X - Y) \leq 3\alpha\sigma_{i_0}^2(u_{i_0}^*)^2 \rightarrow 3\alpha\sigma_{i_0}^2(u_{i_0}^*)^2, \quad (75)$$

which vanishes as $\alpha \downarrow 0$.

Therefore, taking the limit as $\delta \downarrow 0$, we obtain:

$$2\gamma \leq \sum_{j=1}^m K_{i_0j} (U_j(\bar{t}, \bar{x}) - W_j(\bar{t}, \bar{x})) + O(\alpha). \quad (76)$$

Letting $\alpha \downarrow 0$, we define the maximum difference across all regimes and states as:

$$Z_{\max} = U_{i_0}(\bar{t}, \bar{x}) - W_{i_0}(\bar{t}, \bar{x}) = \max_{j \in S} (U_j(\bar{t}, \bar{x}) - W_j(\bar{t}, \bar{x})) > 0. \quad (77)$$

Since K is a Metzler matrix with row-deficiency ($K_{i_0j} \geq 0$ for all $j \neq i_0$ and $\sum_{j=1}^m K_{i_0j} = -\epsilon_{i_0} \leq 0$), we have:

$$\sum_{j=1}^m K_{i_0j} (U_j(\bar{t}, \bar{x}) - W_j(\bar{t}, \bar{x})) \leq K_{i_0i_0} Z_{\max} + \sum_{j \neq i_0} K_{i_0j} Z_{\max} = \left(\sum_{j=1}^m K_{i_0j} \right) Z_{\max} = -\epsilon_{i_0} Z_{\max}. \quad (78)$$

Thus, (76) reduces to:

$$2\gamma \leq -\epsilon_{i_0} Z_{\max} \leq 0, \quad (79)$$

which is a contradiction since $\gamma > 0$ and $Z_{\max} > 0$. This establishes the comparison principle, and uniqueness follows immediately.

B.2 Viscosity Subsolution and Supersolution Verification

We now verify that the candidate vector of value functions $\mathbf{V}(t, x)$ constructed in Section 3 is a viscosity solution.

Stage 1: Viscosity Subsolution Property. Let $i \in \mathcal{S}$ and $(t_0, x_0) \in [0, T) \times (0, \infty)$ be arbitrary. Let $\psi \in C^{1,2}([0, T] \times [0, \infty))$ be a test function such that $V_i - \psi$ achieves a local maximum at (t_0, x_0) with $V_i(t_0, x_0) = \psi(t_0, x_0)$. Because V_i is continuously differentiable in t and twice continuously differentiable in x , the first-order and second-order derivatives must coincide at the local maximum:

$$\frac{\partial V_i}{\partial t}(t_0, x_0) = \frac{\partial \psi}{\partial t}(t_0, x_0), \quad \frac{\partial V_i}{\partial x}(t_0, x_0) = \frac{\partial \psi}{\partial x}(t_0, x_0), \quad \frac{\partial^2 V_i}{\partial x^2}(t_0, x_0) \leq \frac{\partial^2 \psi}{\partial x^2}(t_0, x_0). \quad (80)$$

We evaluate the obstacle term: since the investor can liquidate the asset immediately, the value function satisfies $V_i(t_0, x_0) \geq x_0$ everywhere. If $V_i(t_0, x_0) > x_0$, then (t_0, x_0) lies in the continuation domain C_i . In this domain, the ordinary differential equations (22), (23), and (24) are satisfied. Evaluating the continuation operator $\mathcal{H}_i$ at the derivatives of ψ yields:

$$\mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) - V_i(t_0, x_0) + x_0 = \mathcal{H}_i(t_0, x_0, V_i, DV_i, D^2V_i) - V_i(t_0, x_0) + x_0 \geq 0, \quad (81)$$

due to the optimality of u_i^* under the coefficients. If $V_i(t_0, x_0) = x_0$, the immediate liquidation boundary is active, which guarantees:

$$\max \{ \mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) - V_i(t_0, x_0) + x_0, \quad x_0 - V_i(t_0, x_0) \} \geq 0, \quad (82)$$

since the second term is exactly 0. Thus, V_i is a viscosity subsolution.

Stage 2: Viscosity Supersolution Property. Let $\psi \in C^{1,2}([0, T] \times [0, \infty))$ be a test function such that $V_i - \psi$ achieves a local minimum at (t_0, x_0) with $V_i(t_0, x_0) = \psi(t_0, x_0)$. At the local minimum,

we have:

$$\frac{\partial V_i}{\partial t}(t_0, x_0) = \frac{\partial \psi}{\partial t}(t_0, x_0), \quad \frac{\partial V_i}{\partial x}(t_0, x_0) = \frac{\partial \psi}{\partial x}(t_0, x_0), \quad \frac{\partial^2 V_i}{\partial x^2}(t_0, x_0) \geq \frac{\partial^2 \psi}{\partial x^2}(t_0, x_0). \quad (83)$$

We must prove that:

$$\max \{ \mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) - V_i(t_0, x_0) + x_0, \quad x_0 - V_i(t_0, x_0) \} \leq 0, \quad (84)$$

which is equivalent to showing that both terms are non-positive. The second term is non-positive since $V_i(t, x) \geq x$ for all (t, x) , which implies $x_0 - V_i(t_0, x_0) \leq 0$. For the first term, we evaluate the continuation operator. By definition, (t_0, x_0) must lie either in the continuation domain C_i or on the liquidation boundary. In either case, the coefficient system guarantees that:

$$\mathcal{H}_i(t_0, x_0, V_i, DV_i, D^2V_i) - V_i(t_0, x_0) + x_0 \leq 0, \quad (85)$$

with equality holding strictly inside C_i . Since $\frac{\partial^2 V_i}{\partial x^2}(t_0, x_0) \geq \frac{\partial^2 \psi}{\partial x^2}(t_0, x_0)$ and the coefficient of the second derivative is $\frac{1}{2}\sigma_i^2(u_i^*)^2 \geq 0$, we obtain:

$$\mathcal{H}_i(t_0, x_0, V_i, D\psi, D^2\psi) \leq \mathcal{H}_i(t_0, x_0, V_i, DV_i, D^2V_i) \leq V_i(t_0, x_0) - x_0. \quad (86)$$

This establishes the supersolution property.

C Proof of Theorem 3 (Main Unification Theorem)

Stage 1: Verification of the Expected Survival Wealth Function. For any continuous allocation $u(t, x)$ and prior parameter λ , the infinitesimal generator $\mathcal{A}^{u, \lambda}$ of the regime-switching diffusion with absorption acting on a smooth function $g_i(t, x, \lambda)$ is defined as:

$$\mathcal{A}^{u, \lambda} g_i = \frac{\partial g_i}{\partial t} + (rx + u(\mu_i - r - \sigma_i \lambda)) \frac{\partial g_i}{\partial x} + \frac{1}{2} \sigma_i^2 u^2 \frac{\partial^2 g_i}{\partial x^2}. \quad (87)$$

In the continuation domain C_i , the survival-weighted expected terminal wealth $g_i(t, x, \lambda) = \mathbb{E}_{t, x, i}^{\mathbb{Q}^\lambda}[Y_T]$ satisfies the Kolmogorov backward equation:

$$\mathcal{A}^{u^*, \lambda} g_i(t, x, \lambda) + \sum_{j=1}^m K_{ij} g_j(t, x, \lambda) = 0, \quad (88)$$

subject to the terminal boundary condition $g_i(T, x, \lambda) = x$.

We substitute the linear ansatz $g_i(t, x, \lambda) = x d_i(t) + C_i(t) \alpha_i(\lambda)$, where $\alpha_i(\lambda) = \mu_i - r - \sigma_i \lambda$.

Computing the derivatives yields:

$$\frac{\partial g_i}{\partial t} = x d'_i(t) + C'_i(t) \alpha_i(\lambda), \quad \frac{\partial g_i}{\partial x} = d_i(t), \quad \frac{\partial^2 g_i}{\partial x^2} = 0. \quad (89)$$

Substituting these back into the backward equation, we obtain:

$$x d'_i(t) + C'_i(t) \alpha_i(\lambda) + (rx + u_i^*(t)(\mu_i - r - \sigma_i \lambda)) d_i(t) + \sum_{j=1}^m K_{ij} [x d_j(t) + C_j(t) \alpha_j(\lambda)] = 0. \quad (90)$$

Grouping the terms by the state variable x , we have:

$$x \left[d'_i(t) + r d_i(t) + \sum_{j=1}^m K_{ij} d_j(t) \right] + C'_i(t) \alpha_i(\lambda) + u_i^*(t) d_i(t) \alpha_i(\lambda) + \sum_{j=1}^m K_{ij} C_j(t) \alpha_j(\lambda) = 0. \quad (91)$$

Since this equation must hold for all x , the coefficient of x must vanish identically, yielding the ordinary differential equation (22):

$$d'_i(t) + r d_i(t) + \sum_{j=1}^m K_{ij} d_j(t) = 0, \quad (92)$$

with terminal condition $d_i(T) = 1$.

Since the remaining terms are linear in λ , we evaluate at the reference parameter $\lambda = 0$. Using the identity $\alpha_j(\lambda) = \frac{\mu_j - r - \sigma_j \lambda}{\mu_i - r - \sigma_i \lambda} \alpha_i(\lambda)$, evaluating at $\lambda = 0$ yields:

$$C'_i(t) + u_i^*(t) d_i(t) + \sum_{j=1}^m K_{ij} C_j(t) \frac{\mu_j - r}{\mu_i - r} = 0, \quad (93)$$

which is the differential equation (23) with terminal condition $C_i(T) = 0$.

Stage 2: Verification of the Value Function and Optimal Control. We now substitute the linear value function ansatz $V_i(t, x) = x d_i(t) + B_i(t)$ into the coupled HJB equation:

$$\sup_u \left\{ \mathcal{A}^u V_i(t, x) + \sum_{j=1}^m K_{ij} V_j(t, x) - c \sigma_i^2 u^2 d_i^2(t) - \beta \sigma_i^2 \delta_v^2 C_i(t) u d_i(t) - \eta \phi_i \right\} = 0. \quad (94)$$

Calculating the derivatives of the ansatz yields:

$$\frac{\partial V_i}{\partial t} = x d'_i(t) + B'_i(t), \quad \frac{\partial V_i}{\partial x} = d_i(t), \quad \frac{\partial^2 V_i}{\partial x^2} = 0. \quad (95)$$

Substituting these into the HJB equation and grouping by x gives:

$$x \left[d_i'(t) + r d_i(t) + \sum_{j=1}^m K_{ij} d_j(t) \right] + B_i'(t) + \sup_u \mathcal{J}(u) - \eta \phi_i + \sum_{j=1}^m K_{ij} B_j(t) = 0, \quad (96)$$

where the continuous allocation objective function $\mathcal{J}(u)$ is:

$$\mathcal{J}(u) = u(\mu_i - r)d_i(t) - c\sigma_i^2 u^2 d_i^2(t) - \beta\sigma_i^2 \delta_v^2 C_i(t) u d_i(t). \quad (97)$$

The coefficient of x is zero by the ODE for $d_i(t)$. Since $c > 0$, the objective $\mathcal{J}(u)$ is strictly concave with respect to u . Differentiating with respect to u and setting the derivative to zero yields:

$$\frac{\partial \mathcal{J}}{\partial u} = (\mu_i - r)d_i(t) - 2c\sigma_i^2 u d_i^2(t) - \beta\sigma_i^2 \delta_v^2 C_i(t) d_i(t) = 0. \quad (98)$$

Solving for u isolates the unique optimal continuous asset allocation:

$$u_i^*(t) = \frac{\mu_i - r - \beta\sigma_i^2 \delta_v^2 C_i(t)}{2c\sigma_i^2 d_i(t)}. \quad (99)$$

Substituting $u_i^*(t)$ back into the HJB equation, the remaining terms define the ODE for $B_i(t)$ with terminal condition $B_i(T) = 0$:

$$B_i'(t) + u_i^*(t)(\mu_i - r)d_i(t) - c\sigma_i^2 (u_i^*(t))^2 d_i^2(t) - \beta\sigma_i^2 \delta_v^2 C_i(t) u_i^*(t) d_i(t) - \eta \phi_i + \sum_{j=1}^m K_{ij} B_j(t) = 0. \quad (100)$$

Finally, the optimal stopping boundary is determined by the condition $V_i(t, B_i^*(t)) = B_i^*(t)$. Substituting the ansatz yields:

$$B_i^*(t) d_i(t) + B_i(t) = B_i^*(t) \implies B_i^*(t)(1 - d_i(t)) = B_i(t) \implies B_i^*(t) = \frac{B_i(t)}{1 - d_i(t)}. \quad (101)$$

This completes the proof.

D Proof of Theorem 4 (Boundary Degeneration Theorem)

As the structural spectral gap of the negated generator closes ($\alpha(K) \downarrow 0$), the dominant eigenvalue of the residence operator diverges. The residence matrix is $N = (-K)^{-1}$. The row sums of N represent the expected lifetimes spent in the continuation domain prior to ruin:

$$\mathbf{N}\mathbf{1} = -K^{-1}\mathbf{1}. \quad (102)$$

As $\alpha(K) \downarrow 0$, the operator norm satisfies $\|K^{-1}\| \rightarrow \infty$. Under the Perron–Frobenius theorem, because $-K$ is an M-matrix, the dominant eigenvalue's eigenvector is strictly positive. Consequently, the residence times N_{ij} diverge to infinity for all states in the network components.

Now consider the coefficient $B_i(t)$ solving (25). The terminal condition is $B_i(T) = 0$. Since $\eta > 0$ and the structural fragility penalty rate $\phi_i > 0$, the equation features a negative source term $-\eta\phi_i$. We integrate $B_i(t)$ backward:

$$B_i(t) = \mathbb{E}_{t,i}^{\mathbb{P}} \left[\int_t^T \left(u_{S_s}^* (\mu_{S_s} - r) ds - c\sigma_{S_s}^2 (u_{S_s}^*)^2 d_{S_s}^2 - \beta\sigma_{S_s}^2 \delta_v^2 C_{S_s} u_{S_s}^* ds - \eta\phi_{S_s} \right) ds \right]. \quad (103)$$

The expected cumulative residence time spent in each state diverges as the survival duration expands. The integral of the constant penalty rate $-\eta\phi_i$ is proportional to the expected residence time:

$$\mathbb{E}_{t,i}^{\mathbb{P}} \left[\int_t^T \eta\phi_{S_s} ds \right] \rightarrow \infty \quad \text{as} \quad \alpha(K) \downarrow 0. \quad (104)$$

Because the utility is bounded above by the optimal allocations, the negative ruin penalty dominates. This forces $B_i(t) \rightarrow -\infty$ for all $t < T$. However, the immediate liquidation value is x . The value function satisfies $V_i(t, x) = \max\{xd_i(t) + B_i(t), x\}$. Since $B_i(t) \rightarrow -\infty$, the continuation value is strictly less than the liquidation value for any wealth level $x > 0$. The continuation region:

$$C_i = \{(t, x) : xd_i(t) + B_i(t) > x\} = \{(t, x) : x < B_i^*(t)\}, \quad (105)$$

collapses to the empty set under the Hausdorff metric because the threshold $B_i^*(t) = \frac{B_i(t)}{1-d_i(t)}$ is negative (meaning the stopping condition $x \geq B_i^*(t)$ is satisfied for all $x \geq 0$). Thus, $\lim_{\alpha(K) \downarrow 0} B_i^*(t) = 0$. This completes the proof.

E Proof of Theorem 5 (Residence-Time Localization Theorem)

The non-asymptotic structural ruin probability $\mathbb{P}(\text{ruin})$ is defined by the absorption time of the sub-stochastic regime process S_t . Specifically, the transition kinetics of the latent regime process are entirely independent of the wealth process X_t and its parameters, because the transitions are governed by the killed generator $K = \text{diag}(1 - \epsilon_i)A - I$, which is determined by the transition matrix A and the hazard leakages ϵ_i .

Since the local volatility parameters σ_i are parameters of the wealth diffusion process and do not enter the generator matrix K or the state transition rates of the regime network, we have:

$$\frac{\partial K_{jk}}{\partial \sigma_i} \equiv 0, \quad \forall i, j, k \in \mathcal{S}. \quad (106)$$

Consequently, the residence matrix $N = -K^{-1}$ is independent of σ , yielding:

$$\frac{\partial \mathbb{P}(\text{ruin})}{\partial \sigma_i} \equiv 0. \quad (107)$$

On the other hand, the expected residence matrix entries N_{ii} measure the expected duration before ruin. A larger diagonal entry N_{ii} indicates that the system is expected to spend more time inside the active network prior to collapse. The structural ruin probability over a fixed calendar horizon T_h is bounded by:

$$\mathbb{P}(\text{ruin}) \leq 1 - \exp\left(-T_h \frac{\pi^\top N \mathbf{q}}{\pi^\top N \mathbf{1}}\right). \quad (108)$$

Differentiating this upper bound with respect to N_{ii} yields a strictly positive derivative, since the survival probability is linked to the network survival eigenvalues which scale with the residence matrix coefficients. Thus, network persistence sensitivities are strictly positive. This completes the proof.

F Proof of Theorem 6 (Contraction Mapping, Fixed-Point Convergence, and Bifurcation Analysis)

We develop the full proof of Theorem 6 in two parts: first, by establishing the contraction property of the feedback operator under small sensitivities, and second, by deriving the saddle-node bifurcation threshold and showing the discontinuous collapse of the equilibrium.

F.1 Proof of Contraction Mapping and Fixed-Point Convergence

Let $\mathcal{U} = \{\mathbf{u} \in C([0, T]; \mathbb{R}^m) : 0 \leq u_i(t) \leq U_{\max}, \forall i \in \mathcal{S}, t \in [0, T]\}$ be the closed subset of the Banach space of continuous functions mapping into $\mathbb{R}^m$, equipped with the supremum norm:

$$\|\mathbf{u}\| = \max_{i \in \mathcal{S}} \sup_{t \in [0, T]} |u_i(t)|. \quad (109)$$

The feedback operator $\mathcal{T} : \mathcal{U} \rightarrow \mathcal{U}$ is defined component-wise by:

$$[\mathcal{T}(\mathbf{u})]_i(t) = \frac{\mu_i - r - \beta \sigma_i^2 \delta_v^2 C_i(t; K(\mathbf{u}))}{2c \sigma_i^2 d_i(t; K(\mathbf{u}))}. \quad (110)$$

We show that $\mathcal{T}$ is a contraction mapping by analyzing the sensitivity of the coefficient functions $d_i(t)$ and $C_i(t)$ to changes in the control vector $\mathbf{u}$.

Step 1: Pointwise bound on the generator difference. The generator matrix $K(\mathbf{u})(t)$ is given by $K_{ij}(\mathbf{u})(t) = (1 - \epsilon_i(u_i(t)))A_{ij} - \delta_{ij}$, where $\epsilon_i(u_i) = \epsilon_i^{(0)} + \gamma_i u_i^2$. For any two controls $\mathbf{u}, \mathbf{w} \in \mathcal{U}$, we have:

$$|K_{ij}(\mathbf{u})(t) - K_{ij}(\mathbf{w})(t)| = A_{ij}\gamma_i|u_i(t)^2 - w_i(t)^2| \leq A_{ij}\gamma_i(u_i(t) + w_i(t))|u_i(t) - w_i(t)|. \quad (111)$$

Since $\mathbf{u}, \mathbf{w} \in \mathcal{U}$, we have $u_i(t) + w_i(t) \leq 2U_{\max}$, which yields the pointwise bound:

$$|K_{ij}(\mathbf{u})(t) - K_{ij}(\mathbf{w})(t)| \leq 2\gamma_i A_{ij} U_{\max} \|\mathbf{u} - \mathbf{w}\|. \quad (112)$$

Step 2: Perturbation analysis for the wealth discount factors $d_i(t)$. The wealth discount vector $\mathbf{d}(t)$ solves the linear system $\mathbf{d}'(t) + (rI + K(t))\mathbf{d}(t) = 0$ with terminal condition $\mathbf{d}(T) = \mathbf{1}$. Let $\mathbf{d}_u(t)$ and $\mathbf{d}_w(t)$ be the solutions corresponding to $\mathbf{u}$ and $\mathbf{w}$. The difference vector $\Delta\mathbf{d}(t) = \mathbf{d}_u(t) - \mathbf{d}_w(t)$ satisfies:

$$\Delta\mathbf{d}'(t) + (rI + K_u(t))\Delta\mathbf{d}(t) + (K_u(t) - K_w(t))\mathbf{d}_w(t) = \mathbf{0}, \quad \Delta\mathbf{d}(T) = \mathbf{0}. \quad (113)$$

Integrating this linear system backward from T yields:

$$\Delta\mathbf{d}(t) = \int_t^T e^{r(s-t)} \Phi_{K_u}(s, t) (K_u(s) - K_w(s)) \mathbf{d}_w(s) ds, \quad (114)$$

where $\Phi_{K_u}(s, t)$ is the transition propagator matrix associated with the sub-stochastic generator $K_u(t)$. Since $K_u(t)$ is sub-stochastic, the propagator is a contraction in the supremum norm: $\|\Phi_{K_u}(s, t)\|_{\infty} \leq 1$. Additionally, because $\mathbf{d}_w(s)$ represents survival probabilities under discounting, we have $d_{w,i}(s) \leq 1$ for all s . Taking the supremum norm of both sides:

$$\|\Delta\mathbf{d}(t)\|_{\infty} \leq \int_t^T \|K_u(s) - K_w(s)\|_{\infty} \|\mathbf{d}_w(s)\|_{\infty} ds \leq 2TU_{\max} \left(\max_k \gamma_k \right) \|\mathbf{u} - \mathbf{w}\|. \quad (115)$$

Step 3: Perturbation analysis for the duration coefficients $C_i(t)$. The duration coefficients $\mathbf{C}(t)$ satisfy the linear system $\mathbf{C}'(t) + \mathbf{f}(t) + \tilde{K}(t)\mathbf{C}(t) = 0$ with terminal condition $\mathbf{C}(T) = \mathbf{0}$, where $\tilde{K}_{ij}(t) = K_{ij}(t) \frac{\mu_j - r}{\mu_i - r}$ and $f_i(t) = u_i(t)d_i(t)$. The difference vector $\Delta\mathbf{C}(t) = \mathbf{C}_u(t) - \mathbf{C}_w(t)$ satisfies:

$$\Delta\mathbf{C}'(t) + \tilde{K}_u(t)\Delta\mathbf{C}(t) + (\tilde{K}_u(t) - \tilde{K}_w(t))\mathbf{C}_w(t) + \mathbf{f}_u(t) - \mathbf{f}_w(t) = \mathbf{0}, \quad \Delta\mathbf{C}(T) = \mathbf{0}. \quad (116)$$

Integrating backward:

$$\Delta\mathbf{C}(t) = \int_t^T \Phi_{\tilde{K}_u}(s, t) [(\tilde{K}_u(s) - \tilde{K}_w(s))\mathbf{C}_w(s) + \mathbf{f}_u(s) - \mathbf{f}_w(s)] ds. \quad (117)$$

We bound the terms inside the integral:

1. ****Source term difference****:

$$|f_{u,i}(s) - f_{w,i}(s)| \leq |u_i(s) - w_i(s)|d_{u,i}(s) + w_i(s)|d_{u,i}(s) - d_{w,i}(s)| \leq \|\mathbf{u} - \mathbf{w}\| + U_{\max} \|\Delta \mathbf{d}\|_{\infty}. \quad (118)$$

Substituting (115) gives:

$$\|\mathbf{f}_u - \mathbf{f}_w\|_{\infty} \leq \left(1 + 2TU_{\max}^2 \max_k \gamma_k\right) \|\mathbf{u} - \mathbf{w}\|. \quad (119)$$

2. ****Generator term difference****: Let $C_{w,j}(s) \leq C_{\max}$. Then:

$$|(\tilde{K}_{u,ij} - \tilde{K}_{w,ij})C_{w,j}(s)| \leq 2\gamma_i A_{ij} U_{\max} C_{\max} \frac{\mu_j - r}{\mu_i - r} \|\mathbf{u} - \mathbf{w}\|. \quad (120)$$

Summing over j and taking the supremum norm yields:

$$\|(\tilde{K}_u - \tilde{K}_w)\mathbf{C}_w\|_{\infty} \leq 2U_{\max} C_{\max} H_R \left(\max_k \gamma_k\right) \|\mathbf{u} - \mathbf{w}\|, \quad (121)$$

where $H_R = \max_i \sum_{j=1}^m A_{ij} \frac{\mu_j - r}{\mu_i - r}$.

Integrating these bounds over $[t, T]$ yields:

$$\|\Delta \mathbf{C}(t)\|_{\infty} \leq T \|\Phi_{\tilde{K}_u}\|_{\infty} \left[1 + 2U_{\max} (TU_{\max} + C_{\max} H_R) \max_k \gamma_k\right] \|\mathbf{u} - \mathbf{w}\|. \quad (122)$$

Step 4: Lipschitz continuity of the operator $\mathcal{T}$. Evaluating the operator difference:

$$\begin{aligned} & |[\mathcal{T}(\mathbf{u})]_i(t) - [\mathcal{T}(\mathbf{w})]_i(t)| \\ & \leq \frac{(\mu_i - r)|d_{u,i}(t) - d_{w,i}(t)| + \beta\sigma_i^2\delta_v^2 [C_{u,i}(t)|d_{u,i}(t) - d_{w,i}(t)| + d_{u,i}(t)|C_{u,i}(t) - C_{w,i}(t)|]}{2c\sigma_i^2 d_{u,i}(t)d_{w,i}(t)}. \end{aligned} \quad (123)$$

Since $d_{u,i}(t) \geq \underline{d} > 0$ for all $t \in [0, T]$ (by sub-stochasticity and bounded hazard leakages), the denominator is strictly positive. Substituting the bounds (115) and (122) yields:

$$\|\mathcal{T}(\mathbf{u}) - \mathcal{T}(\mathbf{w})\| \leq L \|\mathbf{u} - \mathbf{w}\|, \quad (124)$$

where the Lipschitz constant is of the form:

$$L = L_1 \max_k \gamma_k + L_2 T, \quad (125)$$

for positive constants L_1, L_2 depending on the system parameters. For sufficiently small feedback sensitivities γ_i and/or small horizon T , we obtain $L < 1$. Under this condition, the operator $\mathcal{T}$ is a contraction mapping on the Banach space, and by the Banach Fixed-Point Theorem, there exists a unique general equilibrium $(\mathbf{u}^*, \mathbf{B}^*, K^*)$.

F.2 Bifurcation and Collapse Analysis

To analyze the bifurcation and collapse behaviors when the feedback sensitivity γ_i is large, we study the algebraic consistency equation for the optimal allocation u_i^* at a given time t :

$$2c\sigma_i^2 d_i(t; \epsilon_i) u_i^* = \mu_i - r - \beta\sigma_i^2 \delta_v^2 C_i(t; \epsilon_i), \quad (126)$$

where $\epsilon_i(u_i^*) = \epsilon_i^{(0)} + \gamma_i(u_i^*)^2$. For a constant allocation u_i , we integrate the backward ODEs for the coefficients:

$$d_i(t; \epsilon_i) = d_i^{(0)}(t) \exp\left(-\gamma_i u_i^2 (T - t)\right), \quad (127)$$

$$C_i(t; \epsilon_i) = u_i d_i(t; \epsilon_i) \left(\frac{1 - e^{-r(T-t)}}{r} \right). \quad (128)$$

Substituting these expressions back into the consistency equation (126) and dividing by $d_i(t; \epsilon_i) > 0$ yields:

$$2c\sigma_i^2 u_i + \beta\sigma_i^2 \delta_v^2 u_i \left(\frac{1 - e^{-r(T-t)}}{r} \right) = \frac{\mu_i - r}{d_i^{(0)}(t)} \exp\left(\gamma_i u_i^2 (T - t)\right). \quad (129)$$

We define the structural constants:

$$A = 2c\sigma_i^2 + \beta\sigma_i^2 \delta_v^2 \left(\frac{1 - e^{-r(T-t)}}{r} \right) > 0, \quad B = \frac{\mu_i - r}{d_i^{(0)}(t)} > 0, \quad (130)$$

so that the equation simplifies to:

$$A u_i \exp\left(-\gamma_i (T - t) u_i^2\right) = B. \quad (131)$$

Let $\theta = \gamma_i (T - t) > 0$ and $g(u) = A u e^{-\theta u^2}$. To find the maximum of $g(u)$ on $(0, \infty)$, we compute the derivative:

$$g'(u) = A e^{-\theta u^2} (1 - 2\theta u^2) = 0 \implies u_{\max} = \frac{1}{\sqrt{2\theta}}. \quad (132)$$

The maximum value of the function is:

$$g(u_{\max}) = A \frac{1}{\sqrt{2\theta}} e^{-1/2} = A \frac{1}{\sqrt{2e\theta}} = A \frac{1}{\sqrt{2e\gamma_i(T-t)}}. \quad (133)$$

A real-valued solution to the consistency equation exists if and only if the maximum value is at least B :

$$B \leq g(u_{\max}) = A \frac{1}{\sqrt{2e\gamma_i(T-t)}} \implies \gamma_i \leq \gamma_{\text{crit}}^i \equiv \frac{A^2}{2e(T-t)B^2}. \quad (134)$$

Substituting A and B back into the threshold expression isolates the critical feedback sensitivity:

$$\gamma_{\text{crit}}^i = \frac{\left(2c\sigma_i^2 + \beta\sigma_i^2\delta_v^2\left(\frac{1-e^{-r(T-t)}}{r}\right)\right)^2}{2e(T-t)\left(\frac{\mu_i-r}{d_i^{(0)}(t)}\right)^2}. \quad (135)$$

When the feedback sensitivity γ_i lies below γ_{crit}^i , there are exactly two solutions to the consistency equation: a stable, high-participation equilibrium u_i^* (where $g'(u^*) > 0$), and an unstable coordination threshold equilibrium (where $g'(u^*) < 0$).

As $\gamma_i \uparrow \gamma_{\text{crit}}^i$, these two solutions converge. At the critical threshold $\gamma_i = \gamma_{\text{crit}}^i$, they merge in a saddle-node (fold) bifurcation. For any feedback sensitivity exceeding the threshold ($\gamma_i > \gamma_{\text{crit}}^i$), no real-valued solution to the consistency equation exists. In this regime, the feedback is too strong to sustain a positive investment equilibrium, forcing the optimal allocation to collapse discontinuously to the boundary state $u_i^* = 0$. Since the investment allocation jumps to 0, the wealth discount factor drops, and the liquidation boundary collapses to the origin ($B_i^*(t) = 0$), forcing a complete and irreversible market participation failure. This completes the proof.

G Proof of Theorem 7 and Proposition 1 (Strategic Cascades and Multiplicity)

We construct the complete game-theoretic and order-theoretic proof of the strategic liquidation cascade and the multiplicity of coordination equilibria. The proof is structured in three parts: first, establishing the complete lattice structure and best-response monotonicity; second, proving strategic complementarity via continuation value deflation; and third, characterizing the S-shaped feedback crossings under Tarski's Fixed-Point Theorem.

G.1 Lattice Foundations and Best-Response Monotonicity

Let the stoppage boundary game be played by a continuum of heterogeneous agents indexed by their risk aversion parameter $c \in [\underline{c}, \bar{c}]$. Each investor c chooses a continuous stopping boundary

$B^c \in \mathcal{B}_c$, where $\mathcal{B}_c = C([0, T]; [0, X_{\max}])$ is the set of continuous functions mapping the calendar horizon to wealth thresholds. We equip $\mathcal{B}_c$ with the pointwise partial order:

$$B^1 \geq B^2 \iff B^1(t) \geq B^2(t), \quad \forall t \in [0, T]. \quad (136)$$

Under this pointwise order, the action space $\mathcal{B}_c$ is a complete lattice, meaning that every subset has a supremum and an infimum. The joint strategy space $\mathcal{H} = \prod_{c \in [\underline{c}, \bar{c}]} \mathcal{B}_c$ is the product of complete lattices. By Tychonoff's Theorem, $\mathcal{H}$ equipped with the product order is also a complete lattice.

For a given profile of stopping boundaries $\mathbf{B} = (B^c)_{c \in [\underline{c}, \bar{c}]} \in \mathcal{H}$ and a wealth process trajectory, the optimal stopping time of agent c is defined by:

$$\tau^c(\mathbf{B}) = \inf \{t \geq 0: X_t \leq B^c(t)\}. \quad (137)$$

The aggregate liquidation volume $\mathcal{V}_t: \mathcal{H} \rightarrow [0, 1]$ is given by:

$$\mathcal{V}_t(\mathbf{B}) = \int_{\underline{c}}^{\bar{c}} \mathbb{I}_{\{\tau^c(\mathbf{B}) \leq t\}} dF(c). \quad (138)$$

Let $\mathbf{B}, \mathbf{B}' \in \mathcal{H}$ be two stopping boundary profiles satisfying $\mathbf{B} \geq \mathbf{B}'$ pointwise. For any sample path of the wealth process, since $B^c(t) \geq (B^c)'(t)$ for all t , the stopping times must satisfy $\tau^c(\mathbf{B}) \leq \tau^c(\mathbf{B}')$ almost surely. Consequently, the indicator functions satisfy:

$$\mathbb{I}_{\{\tau^c(\mathbf{B}) \leq t\}} \geq \mathbb{I}_{\{\tau^c(\mathbf{B}') \leq t\}}, \quad \forall t \in [0, T]. \quad (139)$$

Integrating over the distribution of agents, we obtain:

$$\mathcal{V}_t(\mathbf{B}) \geq \mathcal{V}_t(\mathbf{B}'), \quad \forall t \in [0, T]. \quad (140)$$

This proves that the aggregate liquidation volume $\mathcal{V}_t$ is a pointwise monotonically increasing function of the stopping boundary profile $\mathbf{B}$.

G.2 Strategic Complementarity via Continuation Value Deflation

We now prove that a pointwise increase in the aggregate liquidation volume $\mathcal{V}_t$ deflates the expected continuation value of holding the asset.

1. **Impact on Hazard rates:** By the funding market fragility specification (37), the systemic hazard leakage velocities are $\epsilon_i(t) = \epsilon_i^{(0)} + \theta_i \mathcal{V}_t^2$. Pointwise monotonicity of $\mathcal{V}_t$ implies that if $\mathcal{V}_t \geq \mathcal{V}_t'$ pointwise, then $\epsilon_i(t) \geq \epsilon_i'(t)$ pointwise.

2. **Impact on the Generator:** The killed transition generator matrix is $K(t) = \text{diag}(1 - \epsilon_1(t), \dots, 1 - \epsilon_m(t))A - I$. Under the two profiles, the difference matrix is:

$$K(t) - K'(t) = \text{diag} \left(-(\epsilon_1(t) - \epsilon'_1(t)), \dots, -(\epsilon_m(t) - \epsilon'_m(t)) \right) A. \quad (141)$$

Since $\epsilon_i(t) \geq \epsilon'_i(t)$ and $A_{ij} \geq 0$, the matrix $K(t) - K'(t)$ has non-positive diagonal entries and non-positive off-diagonal entries, ensuring $K(t) \leq K'(t)$ element-wise.

3. **Pointwise Comparison of discount factors:** The wealth discount factors $\mathbf{d}(t)$ satisfy the backward linear system $\mathbf{d}'(t) + (rI + K(t))\mathbf{d}(t) = \mathbf{0}$ with $\mathbf{d}(T) = \mathbf{1}$. Since $K(t) \leq K'(t)$ element-wise and both matrices are Metzler, the comparison theorem for linear systems of ordinary differential equations guarantees that the solutions satisfy:

$$d_i(t; K) \leq d_i(t; K'), \quad \forall t \in [0, T], \quad \forall i \in \mathcal{S}. \quad (142)$$

Similarly, the expected duration coefficients $\mathbf{C}(t)$ satisfy $\mathbf{C}'(t) + \mathbf{f}(t) + \tilde{K}(t)\mathbf{C}(t) = \mathbf{0}$, with $f_i(t) = u_i(t)d_i(t)$ and $\tilde{K}_{ij}(t) = K_{ij}(t)\frac{\mu_j - r}{\mu_i - r}$. Since $d_i(t)$ is deflated and the diagonal elements of the generator are more negative, we obtain:

$$C_i(t; K) \leq C_i(t; K'), \quad \forall t \in [0, T], \quad \forall i \in \mathcal{S}. \quad (143)$$

Finally, the continuation value is $V_c(t, x) = xd_i(t) + B_i(t)$. Since the discount factors are pointwise decreasing in $\mathcal{V}_t$, the continuation value satisfies:

$$V_c(t, x; \mathbf{B}) \leq V_c(t, x; \mathbf{B}'), \quad \text{for } \mathbf{B} \geq \mathbf{B}'. \quad (144)$$

The optimal stopping boundary $B^c(t)$ is defined as the wealth level where the continuation value hits the immediate liquidation value x :

$$B^c(t) = \inf \{x \geq 0: V_c(t, x) \leq x\} = \frac{B_i(t)}{1 - d_i(t)}. \quad (145)$$

Since a pointwise increase in $\mathbf{B}^{-c}$ deflates the continuation value $V_c(t, x)$, the wealth level required to satisfy the inequality $V_c(t, x) \leq x$ must increase. This shifts the stopping boundary upward:

$$\mathcal{R}_c(\mathbf{B}^{-c})(t) \geq \mathcal{R}_c((\mathbf{B}^{-c})')(t), \quad \text{for } \mathbf{B}^{-c} \geq (\mathbf{B}^{-c})'. \quad (146)$$

This establishes strategic complementarity: the best-response operator $\mathcal{R}: \mathcal{H} \rightarrow \mathcal{H}$ is a pointwise monotonically increasing (order-preserving) mapping on the complete lattice $\mathcal{H}$.

G.3 Equilibrium Multiplicity and Coordination Tipping

By Tarski's Fixed-Point Theorem, any order-preserving mapping on a non-empty complete lattice has a non-empty, complete lattice of fixed points. Hence, the set of rational expectations equilibria is a complete lattice. In particular, there exist a unique maximal equilibrium boundary profile $\mathbf{B}_{\max}^*$ (representing early liquidation, corresponding to complete market run) and a unique minimal equilibrium boundary profile $\mathbf{B}_{\min}^*$ (representing high participation).

Under Proposition 1, the feedback mapping $f(\mathcal{V}^e): [0, 1] \rightarrow [0, 1]$ represents the actual aggregate liquidation volume given an expected liquidation volume $\mathcal{V}^e$:

$$f(\mathcal{V}^e) = \int_{\underline{c}}^{\bar{c}} \mathbb{P}(\tau^c(K(\mathcal{V}^e), P(\mathcal{V}^e)) \leq t) dF(c). \quad (147)$$

Since the stopping times τ^c are pointwise decreasing in $\mathcal{V}^e$, the probability of stopping before t is strictly increasing in $\mathcal{V}^e$, ensuring $f'(\mathcal{V}^e) > 0$.

If the price impact sensitivity λ_{PI} and the funding market fragility parameter θ_i are small, the feedback is weak, and the mapping f is flat, yielding a unique crossing of the 45-degree line.

Conversely, if the price impact sensitivity exceeds the critical pecuniary threshold ($\lambda_{PI} > \lambda_{\text{crit}}$) and the funding market fragility parameter exceeds the critical solvency threshold ($\theta_i > \theta_{\text{crit}}^i$), the feedback mapping is S-shaped:

1. For low expected liquidations ($\mathcal{V}^e \approx 0$), actual liquidations remain minimal because asset prices are high and hazard rates are low: $f(0) > 0$ but small.
2. When the expected liquidation volume crosses a tipping threshold, the price impact and margin call cascades trigger a non-linear default loop, causing actual liquidations to escalate rapidly: $f'(\mathcal{V}^e) > 1$ in this intermediate range.
3. For high expected liquidations ($\mathcal{V}^e \approx 1$), asset prices collapse and the hazard rates explode, forcing almost all participants to stop early: $f(1) \approx 1$, with the curve flattening due to saturation.

Since the continuous mapping $f(\mathcal{V}^e)$ satisfies $f(0) > 0$, $f(1) \leq 1$, and its derivative exceeds 1 at the tipping point, the Intermediate Value Theorem guarantees that the curve crosses the 45-degree line exactly three times. These crossings correspond to the three rational expectations equilibria:

1. A stable **high-participation equilibrium** $\mathcal{V}_L^*$ at the lowest crossing, characterized by low liquidations and high asset prices.
2. An unstable **coordination threshold equilibrium** $\mathcal{V}_U^*$ at the intermediate crossing, representing the tipping point.

3. A stable **complete liquidation run** $\mathcal{V}_H^* \approx 1$ at the highest crossing, where the market collapses.

This completes the proof.

H Proof of Proposition 2 (Welfare Wedge and Policies)

We analyze the social planner's optimization problem. The planner chooses the stopping boundaries $\mathbf{B}_{SP}(t) = (B_{SP}^c(t))_{c \in [\underline{c}, \bar{c}]}$ to maximize aggregate welfare:

$$\mathcal{W}_0 = \mathbb{E}^{\mathbb{P}} \left[\int_{\underline{c}}^{\bar{c}} \left(X_{\tau^c} P_{\tau^c} \mathbb{I}_{\{\tau^c < \zeta\}} - \eta \int_0^{\tau^c} \phi_{S_s} ds \right) dF(c) \right]. \quad (148)$$

The competitive agent, by contrast, maximizes their individual value function:

$$V_c(0, x) = \mathbb{E}^{\mathbb{P}} \left[X_{\tau^c} P_{\tau^c} \mathbb{I}_{\{\tau^c < \zeta\}} - \eta \int_0^{\tau^c} \phi_{S_s} ds \right], \quad (149)$$

taking the price process P_t and the hazard rates $\epsilon_i(t)$ as exogenously given.

We differentiate the planner's objective with respect to the stopping boundary $B^c(t)$ of a specific agent c . The planner's first-order condition includes two additional terms (the externalities):

$$\begin{aligned} \frac{\partial \mathcal{W}_0}{\partial B^c(t)} &= \frac{\partial V_c}{\partial B^c(t)} \\ &+ \int_{\underline{c}}^{\bar{c}} \mathbb{E}^{\mathbb{P}} \left[X_{\tau^{c'}} \frac{\partial P_{\tau^{c'}}}{\partial \mathcal{V}_t} \frac{\partial \mathcal{V}_t}{\partial B^c(t)} \mathbb{I}_{\{\tau^{c'} < \zeta\}} \right] dF(c') \\ &+ \int_{\underline{c}}^{\bar{c}} \mathbb{E}^{\mathbb{P}} \left[\frac{\partial \mathbb{I}_{\{\tau^{c'} < \zeta\}}}{\partial \epsilon} \frac{\partial \epsilon}{\partial \mathcal{V}_t} \frac{\partial \mathcal{V}_t}{\partial B^c(t)} \right] dF(c'). \end{aligned} \quad (150)$$

Since $\frac{\partial P}{\partial \mathcal{V}} = -\lambda_{PI} P < 0$ and $\frac{\partial \mathbb{I}}{\partial \epsilon} < 0$, both externalities are strictly negative. Because $\frac{\partial \mathcal{V}_t}{\partial B^c(t)} > 0$, an increase in the individual stopping boundary (earlier liquidation) exerts a negative pecuniary externality (depressing prices for others) and a negative fragility externality (increasing the hazard rate and reducing others' survival probability).

At the decentralized equilibrium boundary $B_D^c(t)$, the individual first-order condition is satisfied: $\frac{\partial V_c}{\partial B^c(t)} = 0$. Evaluating the planner's marginal welfare at this point yields:

$$\frac{\partial \mathcal{W}_0}{\partial B^c(t)} \Big|_{B_D^c(t)} < 0. \quad (151)$$

Since the welfare objective is concave in the boundaries, this indicates that the social planner's

optimal boundary is lower:

$$B_{SP}^c(t) < B_D^c(t). \quad (152)$$

This proves Proposition 5.6. The policy results in Proposition 5.7 follow by noting that LLR asset purchases compress the feedback coefficient $\theta_i(1 - \psi)^2$, which bounds the derivative of the feedback mapping below 1, thereby restoring uniqueness and stabilizing the growth state. This completes the proof.

I Proof of Proposition 3 (Comparative Statics)

We analyze the sensitivities of the optimal investment allocation $u_i^*(t)$ and liquidation boundary $B_i^*(t)$.

1. *Investment Allocation Sensitivity:* The optimal continuous asset allocation is given by:

$$u_i^*(t) = \frac{\mu_i - r - \beta\sigma_i^2\delta_v^2 C_i(t)}{2c\sigma_i^2 d_i(t)}. \quad (153)$$

Differentiating with respect to the risk-aversion parameter c :

$$\frac{\partial u_i^*}{\partial c} = -\frac{\mu_i - r - \beta\sigma_i^2\delta_v^2 C_i(t)}{2c^2\sigma_i^2 d_i(t)} = -\frac{u_i^*(t)}{c}. \quad (154)$$

Since $\mu_i > r$ and $C_i(t) \geq 0$, the optimal allocation u_i^* is positive, which guarantees $\frac{\partial u_i^*}{\partial c} < 0$. Differentiating with respect to the ambiguity-aversion parameter β :

$$\frac{\partial u_i^*}{\partial \beta} = -\frac{\delta_v^2 C_i(t)}{2c d_i(t)}. \quad (155)$$

Since $d_i(t) > 0$, $C_i(t) \geq 0$, and $\delta_v^2 > 0$, this derivative is non-positive: $\frac{\partial u_i^*}{\partial \beta} \leq 0$, with strict inequality holding for all $t < T$. Differentiating with respect to the baseline hazard rate $\epsilon_i^{(0)}$: the generator K depends on the hazard rates, with $\frac{\partial K_{ii}}{\partial \epsilon_i} = -1$. By the backward ODE system, a higher hazard leakage velocity reduces the survival discount factors, which deflates the duration coefficient $C_i(t)$ and wealth discount $d_i(t)$. Since the expected residence times scale down, we obtain $\frac{\partial u_i^*}{\partial \epsilon_i^{(0)}} < 0$.

2. *Liquidation Boundary Sensitivity:* The boundary ODE is given by (25). The coefficient $B_i(t)$ integrates the concave utility terms. Since u_i^* is optimized, the envelope theorem implies that the derivative of the value function with respect to preference parameters is determined by the direct

derivatives of the objective function. For risk aversion c :

$$\frac{\partial B_i}{\partial c} = -\mathbb{E}_{t,i}^{\mathbb{P}} \left[\int_t^T \sigma_{S_s}^2 (u_{S_s}^*)^2 dS_s^2 ds \right] < 0. \quad (156)$$

For ambiguity aversion β :

$$\frac{\partial B_i}{\partial \beta} = -\mathbb{E}_{t,i}^{\mathbb{P}} \left[\int_t^T \frac{1}{2} \sigma_{S_s}^2 \delta_v^2 C_{S_s} u_{S_s}^* dS_s ds \right] < 0. \quad (157)$$

For baseline hazard $\epsilon_i^{(0)}$: the direct impact is to reduce the residence times in the continuation domain, which deflates the continuation value $B_i(t)$ through the discount factor. Since the boundary is $B_i^*(t) = \frac{B_i(t)}{1-d_i(t)}$, a decrease in $B_i(t)$ shrinks the boundary $B_i^*(t)$ (moving it closer to 0). Thus, the sensitivities of B_i^* are all strictly negative. This completes the proof.

J Computational Appendix (R Solver Discretization and Convergence)

The R solver computes the backward coefficients $d_i(t)$ and $C_i(t)$ by integrating the linear system backwards from T . Specifically, the ODE system is discretized on the temporal grid using an implicit Euler step:

$$\mathbf{d}^{n-1} = (I - \Delta t(rI + K))^{-1} \mathbf{d}^n, \quad (158)$$

which is unconditionally stable. The boundary coefficients $\mathbf{B}^n$ are resolved in a similar fashion, matching the viscosity solutions.

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Masthead Graphic

The mast head collage is a beautifully arranged photo collage highlighting the magnificent, intricate details of Jugendstil (Art Nouveau) architecture on Alberta iela (Albert Street) in Rīga. The collage features five distinct photographs set against a muted olive-green background, showcasing the expressive sculptures, mythological motifs, and ornate structural elements that define Mikhail Eisenstein's famous designs. The Top Left (Large Panel) captures a high-angled view of a balcony adorned with a black wrought-iron railing. Top Right: presents a tight, vertical close-up of the same elegant female stone bust seen in the top-left panel. The Middle Right shows a wider architectural slice of a building facade and stylized pillars typical of the grand apartment buildings on the street. The Bottom Left (Large Panel) presents a striking, highly detailed shot of a vibrant blue-tiled facade section dominated by an aggressive, roaring lion's head sculpted in relief. The lion appears to emerge from a heavy bed of stylized oak leaves and branches. Directly below, flanked by the brilliant blue tiles, are two profile busts facing inward wearing Egyptian-style headdresses (nemes), adding an eclectic, historicist flair to the Art Nouveau design. Bottom Right: Captures a ground-level view featuring a classic Art Nouveau sphinx sculpture sitting in profile, guarding an entrance. The door frame behind it features characteristic soft, organic curves, and a small dark blue street sign reading "2a" is visible on the wall, identifying this specific location as the iconic building at Alberta iela 2a.

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